Serious 1 - Secure 3-Tier VPC

Evidence Pack - Final

Haileab T. Bekele - AWS, us-east-1 (N. Virginia) - Built 2026-11-09 to 2026-11-10, one step at a time, with screenshots saved as we go.

Repository: https://github.com/Ha113ab/MY_AWS_PROJECTS - folder serious-1-secure-3tier-vpc/

What this is

A secure 3-tier VPC in AWS, built in the AWS Management Console: web tier (public, load balancer), app tier (private, Auto Scaling servers, no public IP), DB tier (private, RDS MySQL, reached only from the app tier on port 3306). Traffic flows internet - ALB - app - DB only. The DB never touches the internet.

Design

Region: us-east-1. VPC: p1-vpc, vpc-08694d42074a60709, IPv4 CIDR 10.0.0.0/16, DNS hostnames and DNS resolution Enabled. AZs: us-east-1a and us-east-1b. Subnets, all /24: p1-web-a 10.0.0.0/24 (1a, auto-assign Yes), p1-web-b 10.0.1.0/24 (1b, Yes), p1-app-a 10.0.10.0/24 (1a, No), p1-app-b 10.0.11.0/24 (1b, No), p1-db-a 10.0.20.0/24 (1a, No), p1-db-b 10.0.21.0/24 (1b, No). Route tables: p1-rt-public rtb-08d24a8878b8af29a (2 web), p1-rt-private rtb-0fc6acae21be36950 (2 app), p1-rt-db rtb-0cdc9fa65334e18a8 (2 db), each local 10.0.0.0/16, plus 0.0.0.0/0 to the internet gateway in p1-rt-public only. Internet gateway: p1-igw, igw-0c83de58a50570... (truncated in screenshot). Security groups, chained: p1-sg-alb sg-00e39906c1d91de7e (HTTP 80 from 0.0.0.0/0), p1-sg-app sg-0dc782d147f0112b (HTTP 80 from p1-sg-alb only), p1-sg-db sg-063e10e3231772a8b (MySQL 3306 from p1-sg-app only). NACLs: p1-nacl-web acl-04426650b8a64f477, p1-nacl-app acl-0ddcc2d0e9855da45, p1-nacl-db acl-054c237a420067341, 2 subnets each. ALB: p1-alb, p1-alb-1016513694.us-east-1.elb.amazonaws.com, HTTP 80, target group p1-tg-app HTTP 80. Auto Scaling: p1-asg-app, Desired 2 / Min 1 / Max 2, launch template p1-lt-app lt-0707482f22b328e38, Amazon Linux 2023, t3.micro, key p1-key, user data installing httpd with a Hello from $(hostname) page.

Final status - ended 2026-11-10, 22:37 CST, at Haileab's direction

Steps 1-9 are complete and verified. Step 10, Part A created RDS p1-db, Available, MySQL Community 8.4.11, db.t4g.micro, 20 GiB gp2, endpoint p1-db.ci5cw8aia5ax.us-east-1.rds.amazonaws.com:3306, AZ us-east-1b. The last successful test, E51, is the end point: the ALB still serves Hello from ip-10-0-10-221.ec2.internal with p1-db present.

Honest limits - do not claim more than this:

1. p1-db is in the WRONG VPC. It was created in the default VPC (172.31.0.0/16, subnet group default-vpc-0a0cff632591d850c), not in p1-vpc, not in p1-db-a/b. See E52/E53. 2. p1-db is on the WRONG security group, the default SG sg-07c41c74bd4e09c2f, not p1-sg-db sg-063e10e3231772a8b. The app tier cannot reach it on 3306 through that group. 3. No SQL query was run from an app server. The app is a static Hello page with no DB code, and the app instances have no SSH/SSM access. Do not claim a database connection test succeeded. 4. No initial database named p1db was created. DB name is blank. 5. Storage autoscaling and Enhanced Monitoring were left Enabled by console defaults, not turned off as planned. 6. The fix - delete p1-db and recreate it in p1-vpc with p1-db-subnet-group, Public access No, and p1-sg-db only - was NOT done, at Haileab's direction to end here. 7. Part C cleanup was NOT done in this session. As of the end of this build, the hourly resources are still up: ALB p1-alb + 2 app servers, NAT gateway nat-01d3c107223c205cf, and RDS p1-db. Delete them to stop the charges. Do not delete the pre-existing Aurora databases database-1 / database-1-instance-1.

Cost

Steps 1-7: $0. NAT Gateway about $0.045/hr while it exists, plus data. ALB about $0.0225/hr plus LCU, 2 x t3.micro about $0.023/hr together. RDS p1-db about $0.02/hr. Total about $0.11/hr while ALB, app servers, NAT and RDS are all up.

How to run the demo (while the resources are still up)

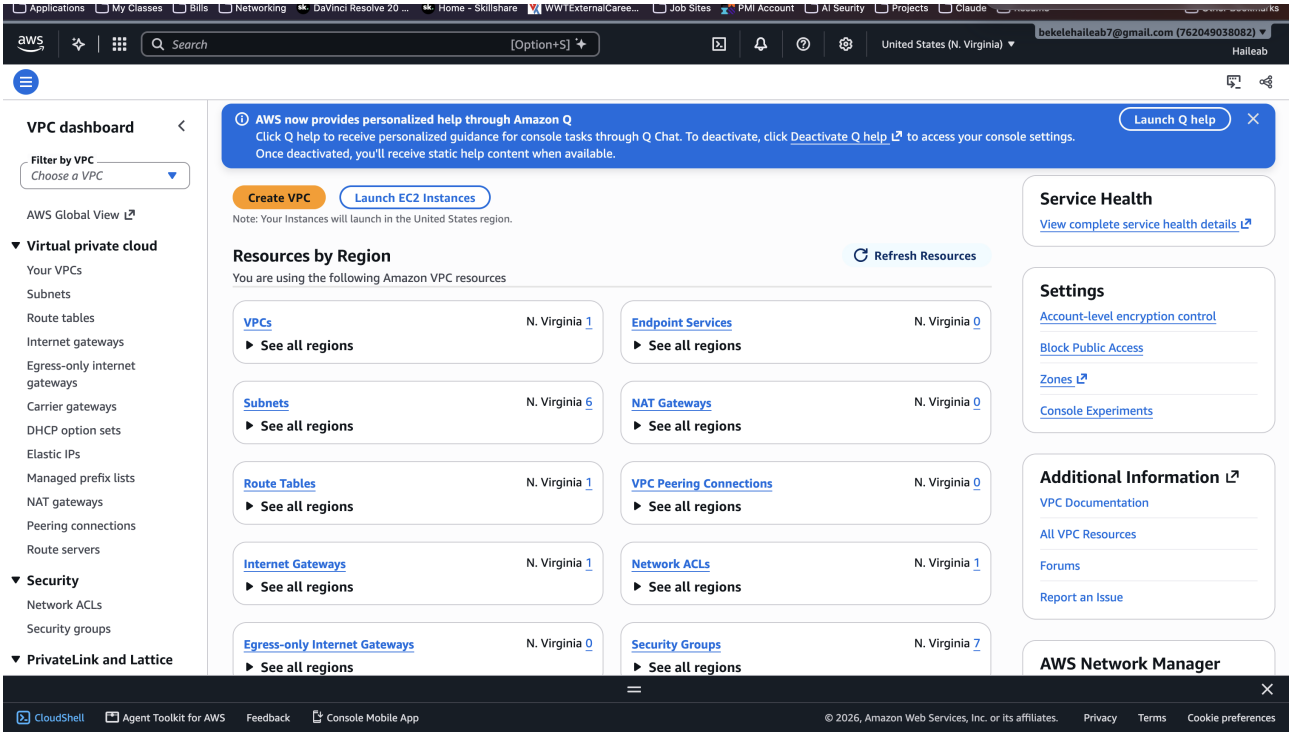
Open http://p1-alb-1016513694.us-east-1.elb.amazonaws.com in a browser. It should show Hello from ip-10-0-10-221.ec2.internal or Hello from ip-10-0-11-245.ec2.internal. Refresh to see the other server, proving load balancing across both AZs.

Evidence index - E01 to E53

Screenshots are in screenshots/ in the repo folder, in the same subfolders as here: vpc-subnets/, security-groups/, load-balancer/, database/. E15, E41, E42 and E43 are pastes, with no screenshot file.

E01 - 2026-11-09 15:46 - vpc-dashboard-us-east-1.png - Not verified - count only

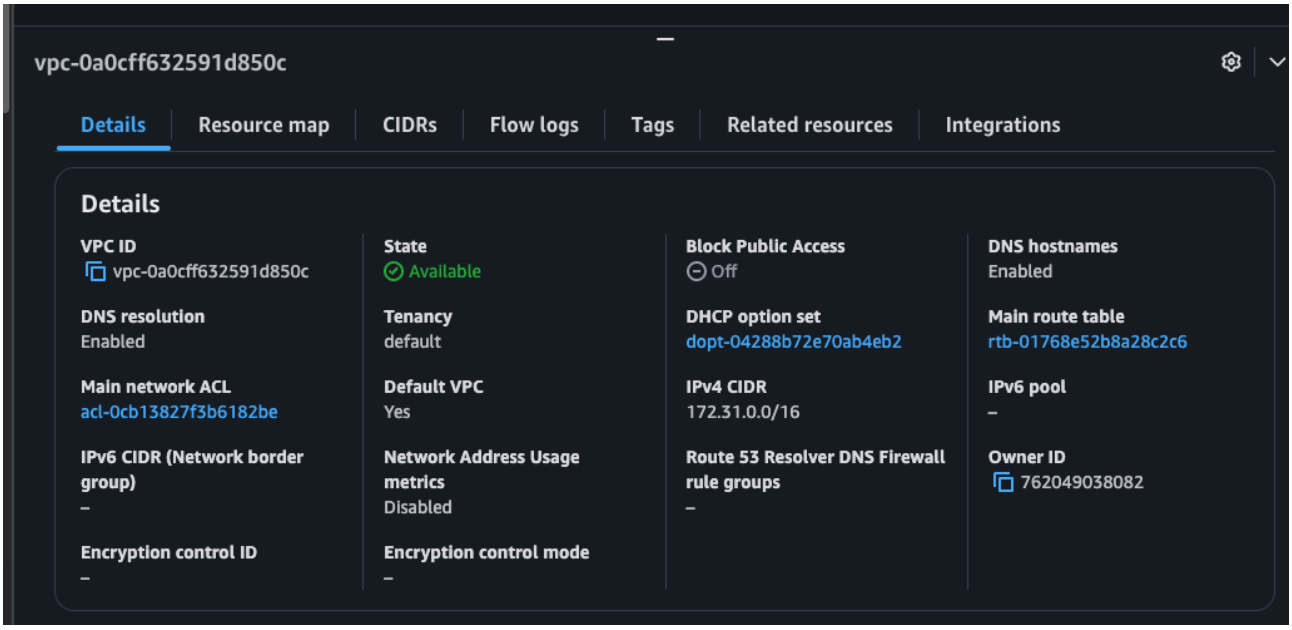
VPC dashboard, us-east-1: 1 VPC, 6 subnets, 1 route table, 1 IGW, 1 NACL, 7 security groups, 0 NAT gateways. A count only, not proof of which VPC.



File: screenshots/vpc-subnets/vpc-dashboard-us-east-1.png

E02 - 2026-11-10 14:20 - your-vpcs-default-vpc.png - Step 1 FAIL - default VPC only

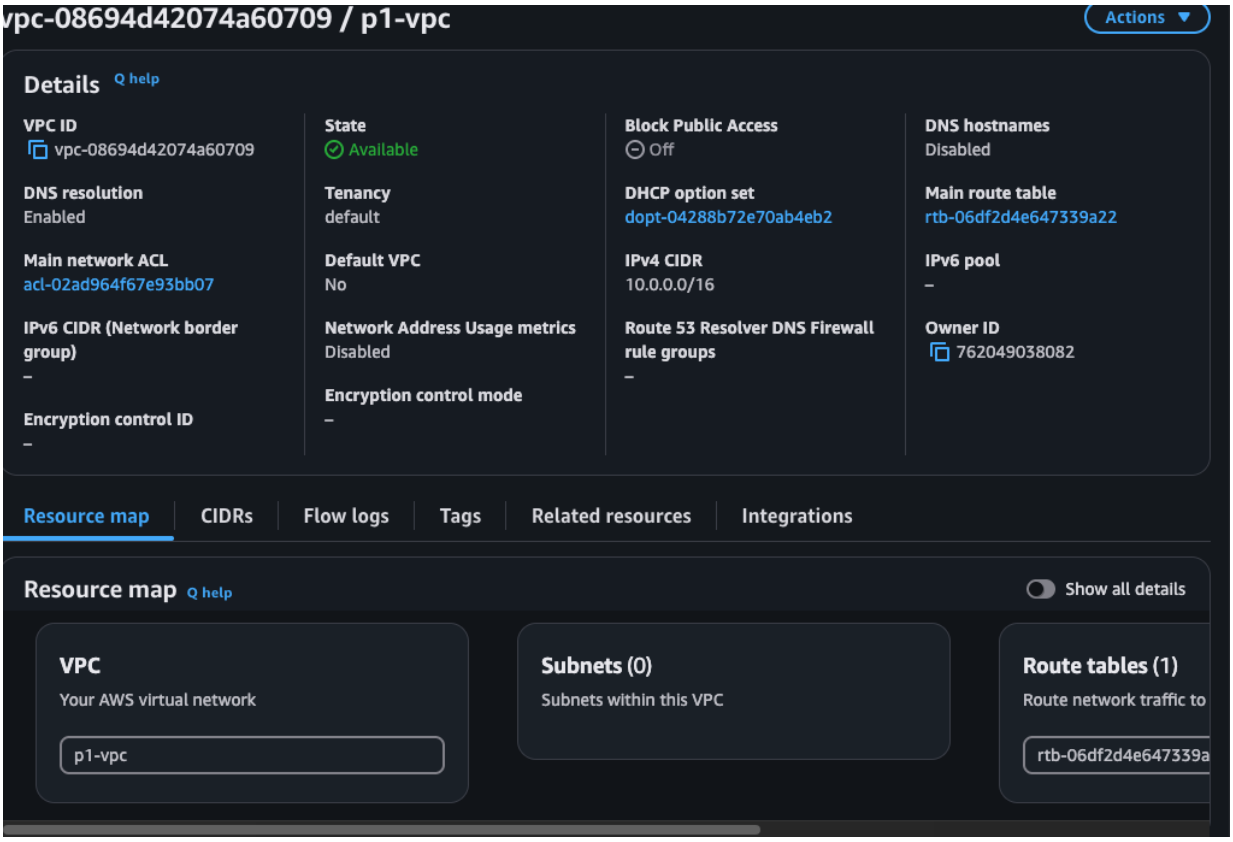
Your VPCs details: only the default VPC, vpc-0a0cff632591d850c, 172.31.0.0/16, Default VPC Yes, DNS hostnames/resolution Enabled.



File: screenshots/vpc-subnets/your-vpcs-default-vpc.png

E03 - 2026-11-10 14:23 - your-vpcs-p1-vpc-created.png - Step 2 complete after fix

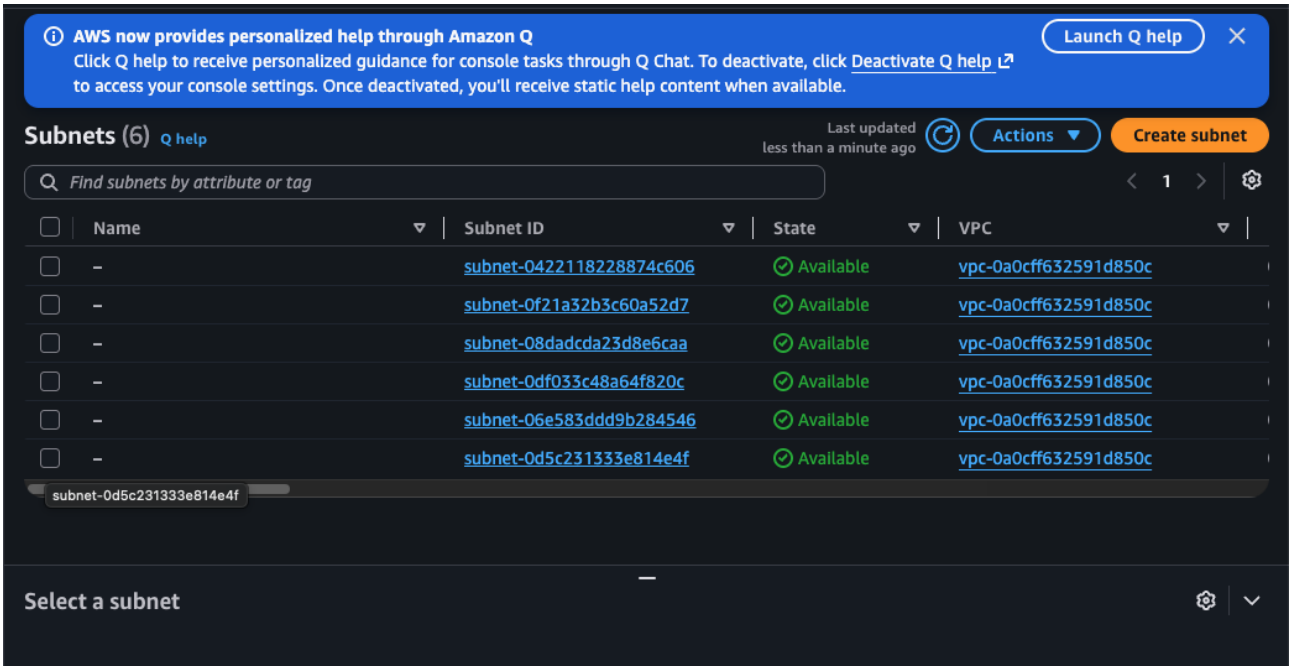
p1-vpc created: vpc-08694d42074a60709, 10.0.0.0/16, Available, Default VPC No. DNS hostnames was Disabled at first, then fixed to Enabled 14:26.



File: screenshots/vpc-subnets/your-vpcs-p1-vpc-created.png

E04 - 2026-11-10 14:29 - subnets-default-vpc.png - Clarification

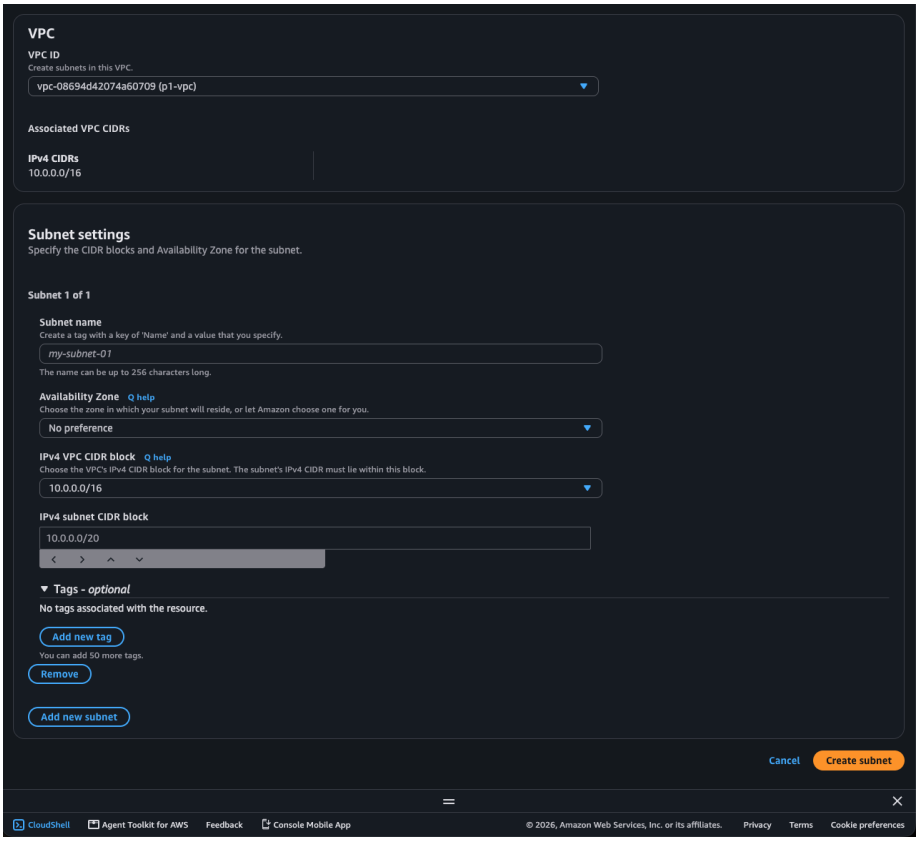
The 6 subnets first seen are the default VPC's, vpc-0a0cff632591d850c, no names. Do not delete, rename or reuse them.



File: screenshots/vpc-subnets/subnets-default-vpc.png

E05 - 2026-11-10 14:31 - create-subnet-form.png - Form trap noted

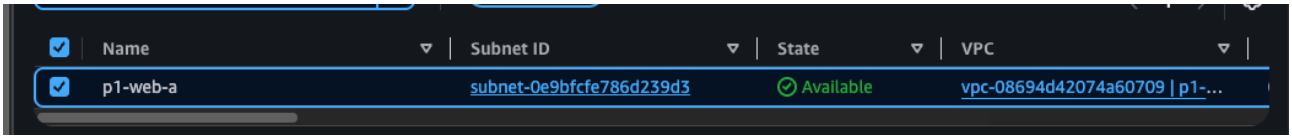
Create subnet form with correct VPC p1-vpc, but defaults wrong: AZ No preference, CIDR 10.0.0.0/20, name empty. Do not create with defaults.



File: screenshots/vpc-subnets/create-subnet-form.png

E06 - 2026-11-10 14:35 - subnet-p1-web-a-created.png - Created, auto-assign fixed later

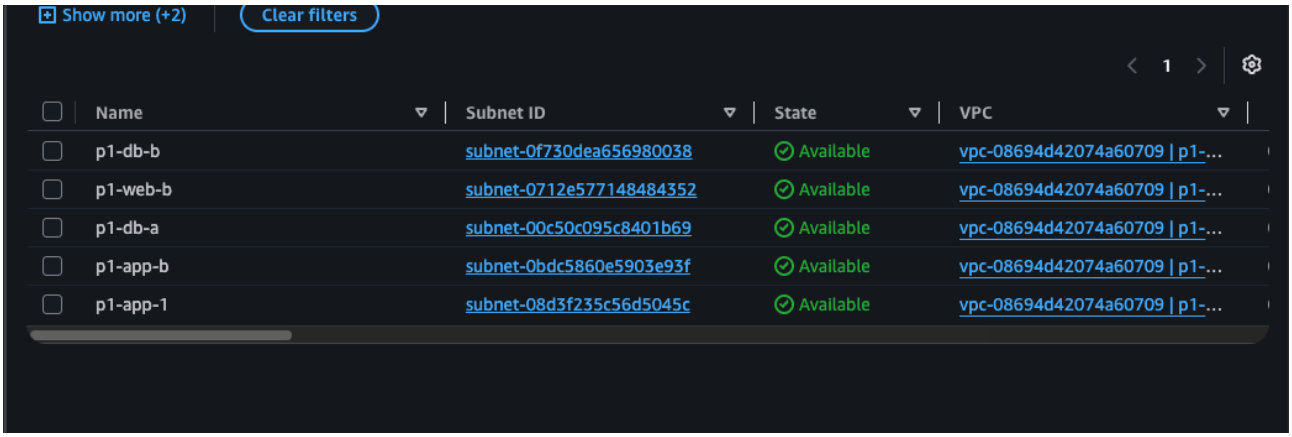
First subnet p1-web-a, subnet-0e9bfcfe786d239d3, 10.0.0.0/24, us-east-1a, Available. Auto-assign public IPv4 was No at that point.



File: screenshots/vpc-subnets/subnet-p1-web-a-created.png

E07 - 2026-11-10 14:43 - subnets-p1-vpc-5-shown.png - Issues found

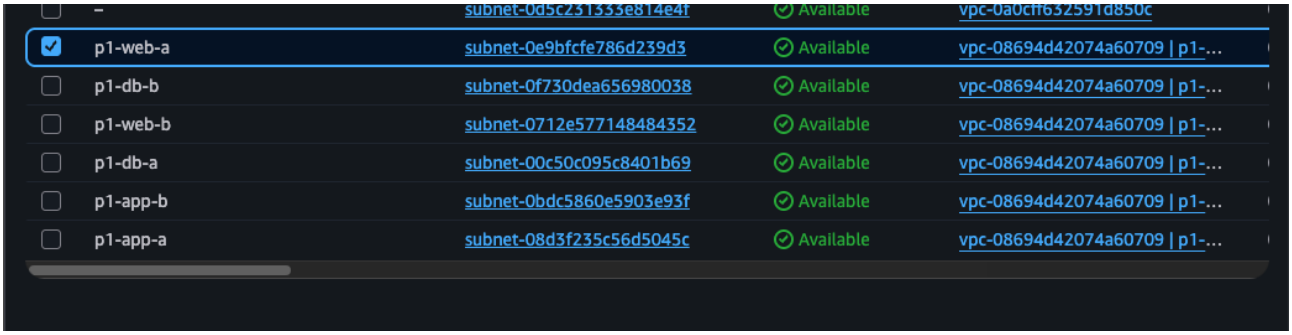
5 subnets in p1-vpc, all Available. Issues: p1-app-1 is a typo for p1-app-a, and p1-web-a is not in this list.



File: screenshots/vpc-subnets/subnets-p1-vpc-5-shown.png

E08 - 2026-11-10 14:45 - subnets-p1-vpc-6-names.png - Partial

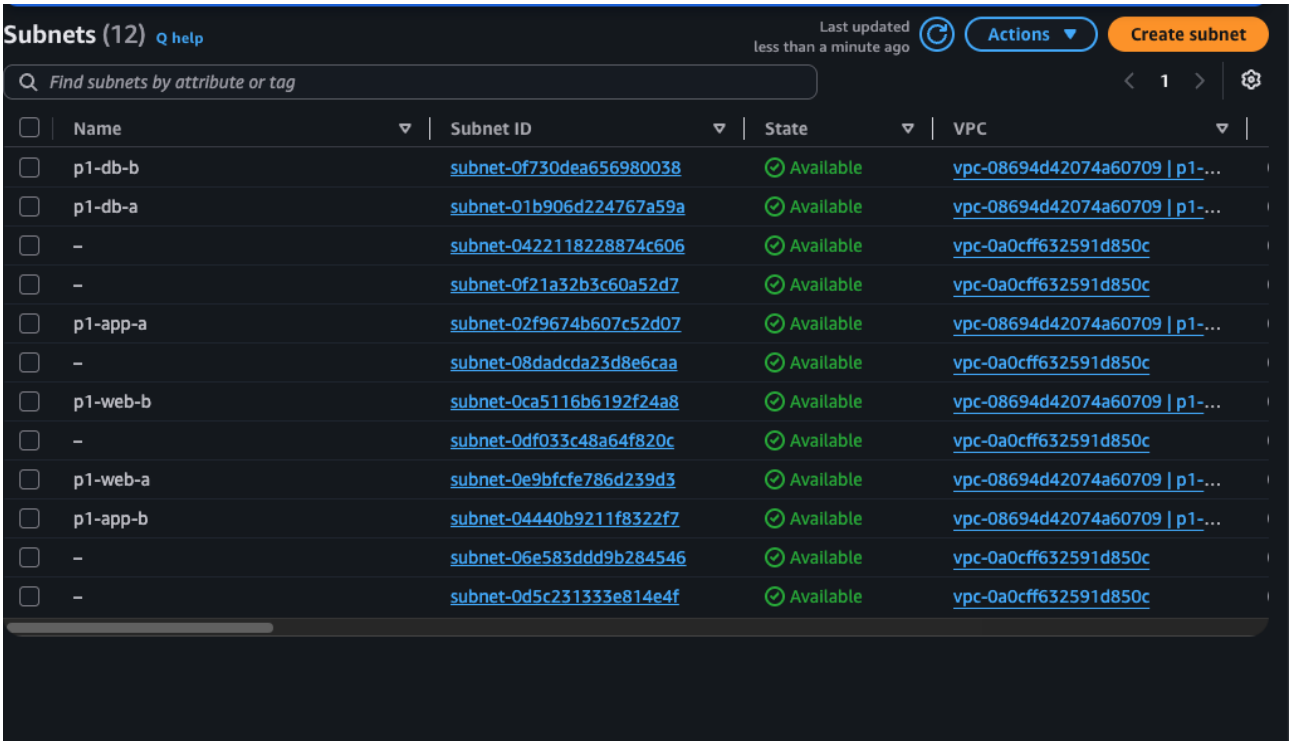
All 6 p1 subnets present, all Available, typo fixed to p1-app-a. CIDRs/AZs and auto-assign not yet verified from this shot.



File: screenshots/vpc-subnets/subnets-p1-vpc-6-names.png

E09 - 2026-11-10 14:57 - subnets-12-total-new-ids.png - Noted

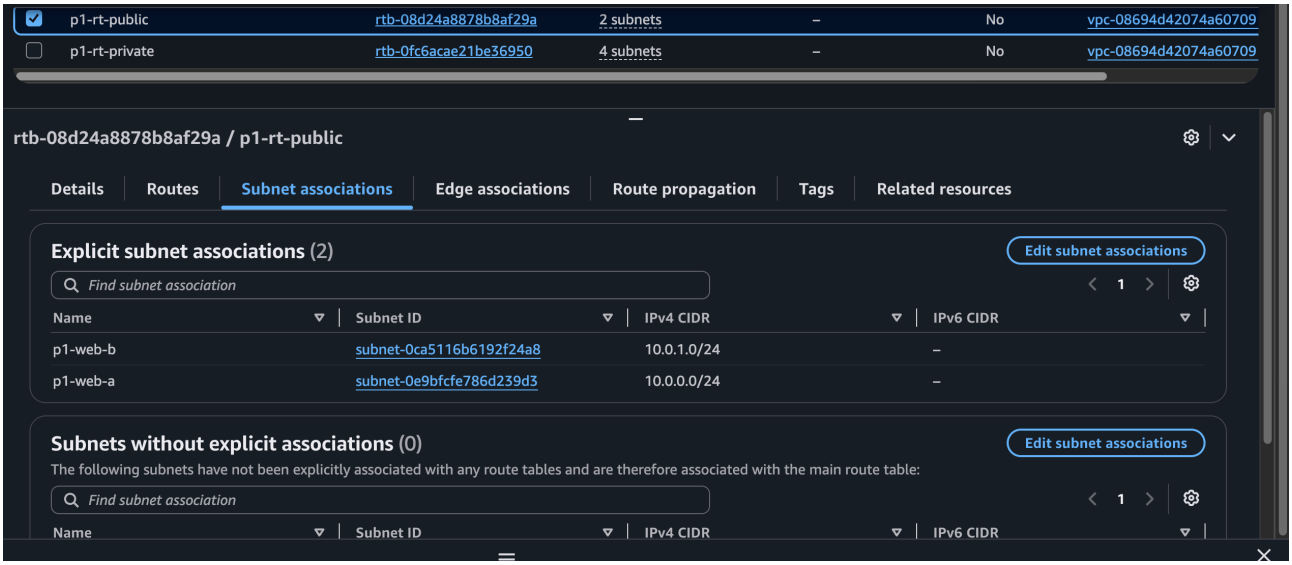
Subnets (12) total, 6 p1 + 6 default. 5 p1 IDs had changed vs 14:45, suggesting delete and recreate.



File: screenshots/vpc-subnets/subnets-12-total-new-ids.png

E10 - 2026-11-10 15:12 - step4-route-tables-p1-rt-public-2-web-subnets.png - Correct

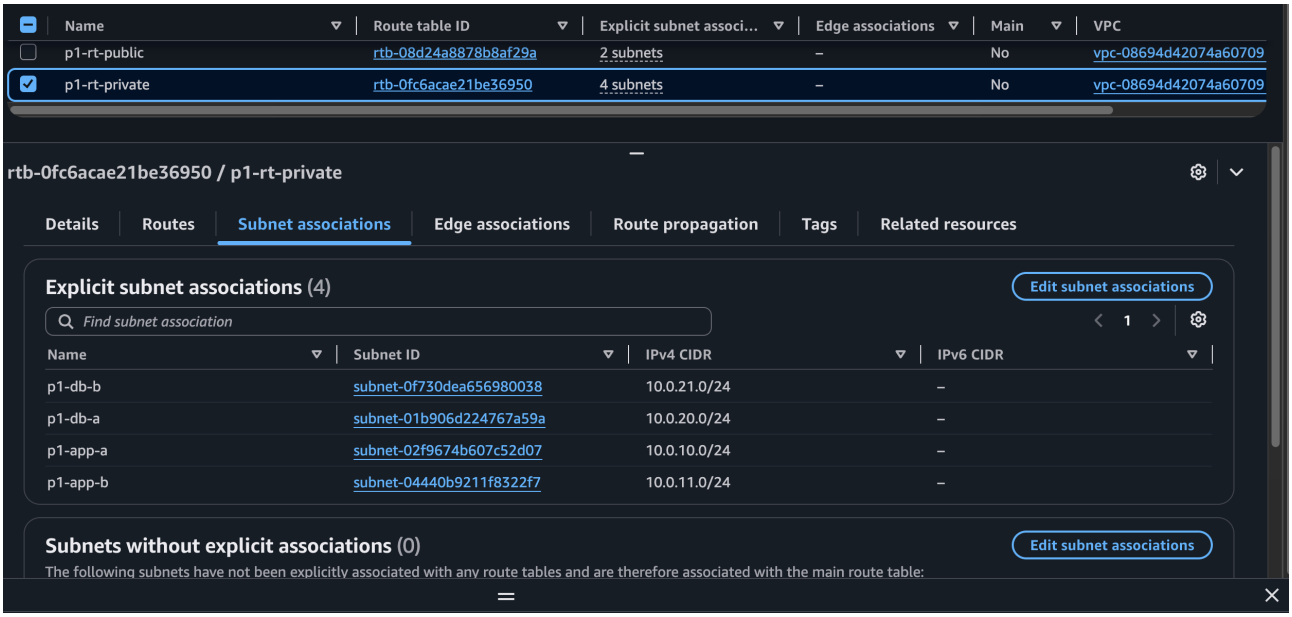
p1-rt-public rtb-08d24a8878b8af29a, explicit associations (2): p1-web-a and p1-web-b.



File: screenshots/vpc-subnets/step4-route-tables-p1-rt-public-2-web-subnets.png

E11 - 2026-11-10 15:11 - step4-route-tables-p1-rt-private-4-subnets-app-and-db.png - Wrong - fixed in E12

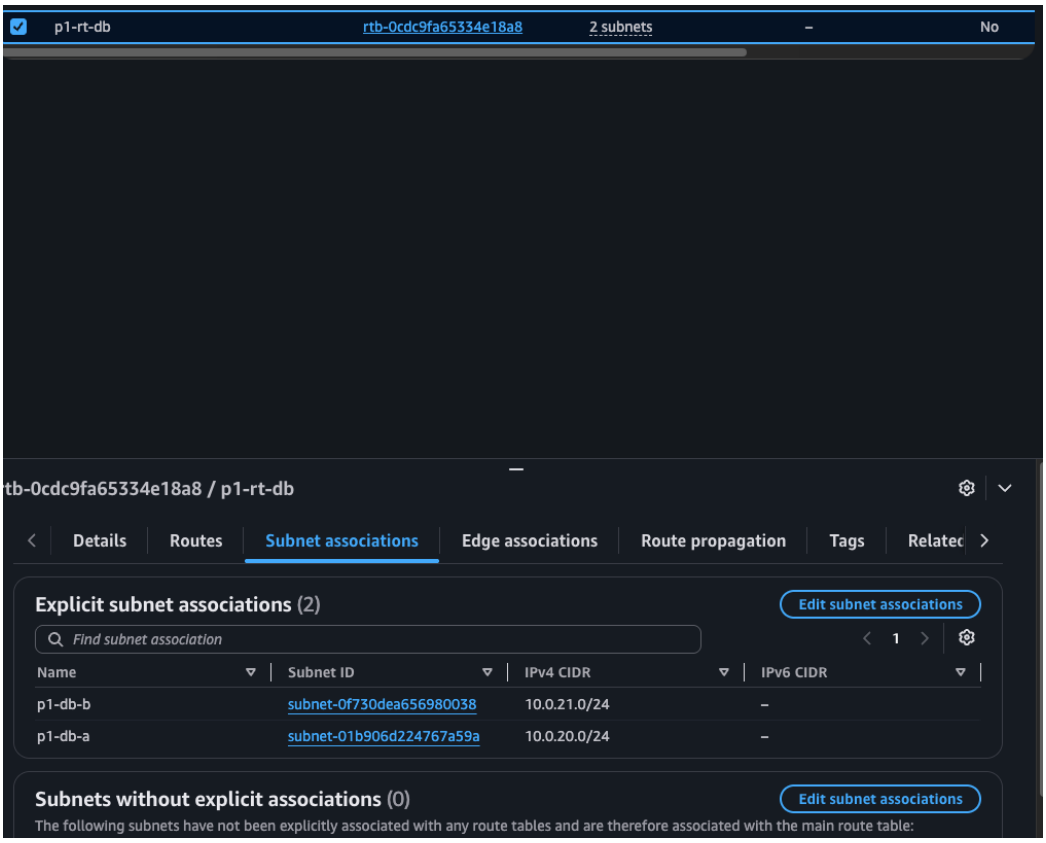
p1-rt-private rtb-0fc6acae21be36950, explicit associations (4): app-a, app-b, db-a, db-b. Wrong, must be 2 app only.



File: screenshots/vpc-subnets/step4-route-tables-p1-rt-private-4-subnets-app-and-db.png

E12 - 2026-11-10 15:19 - step4-route-tables-p1-rt-db-2-db-subnets.png - Step 4 complete

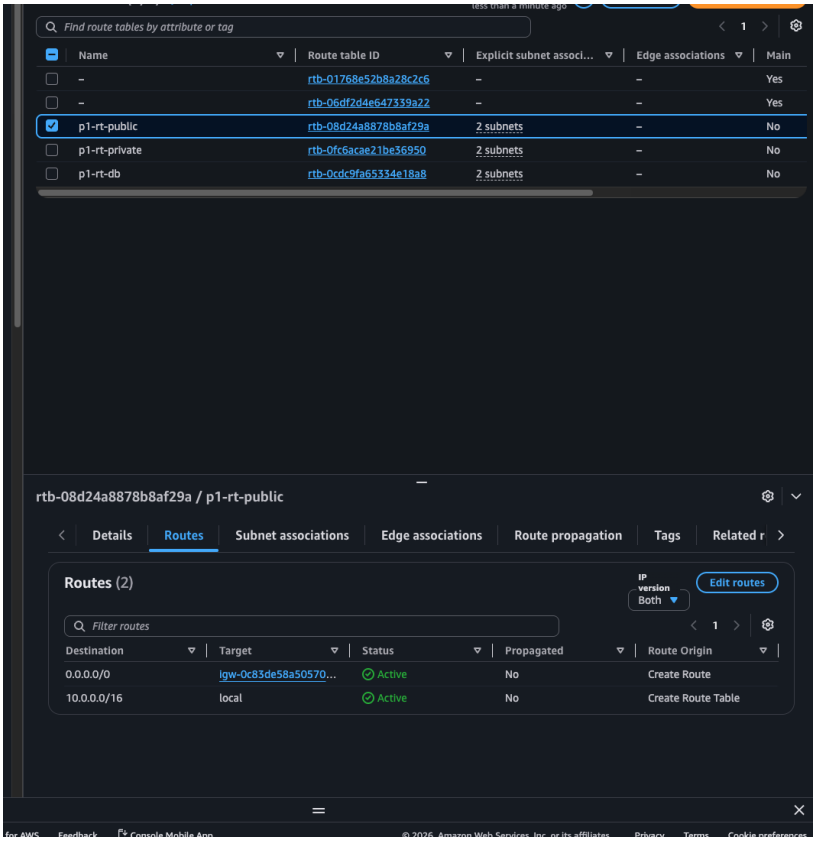
p1-rt-db rtb-0cdc9fa65334e18a8, explicit associations (2): p1-db-a and p1-db-b. Moving DB subnets here removes them from p1-rt-private.



File: screenshots/vpc-subnets/step4-route-tables-p1-rt-db-2-db-subnets.png

E13 - 2026-11-10 15:24 - step5-route-table-p1-rt-public-igw-route.png - Step 5 complete

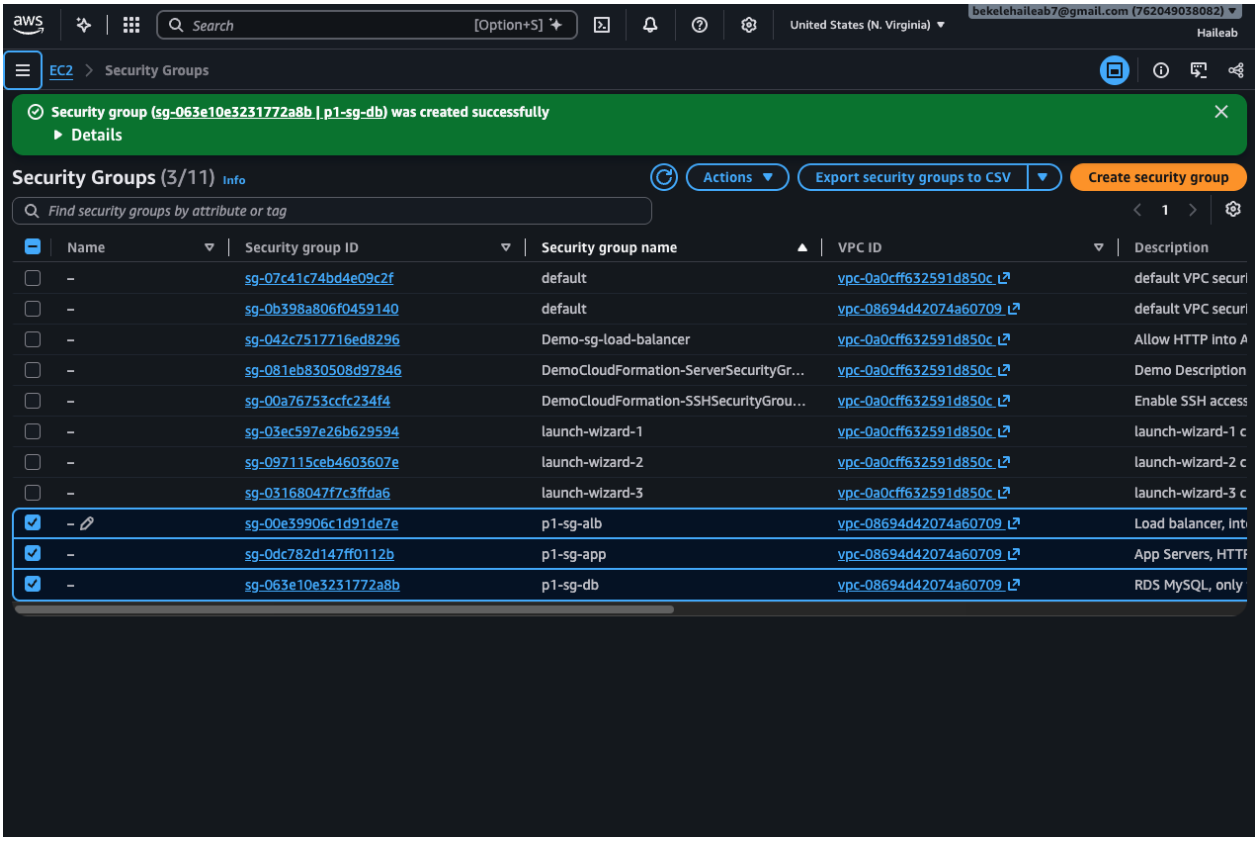
p1-rt-public routes: 0.0.0.0/0 to igw-0c83de58a50570... Active, 10.0.0.0/16 local Active. New IGW, not the default VPC igw-01644b6331e6fef1b. Private 2 subnets, DB 2 subnets confirmed.



File: screenshots/vpc-subnets/step5-route-table-p1-rt-public-igw-route.png

E14 - 2026-11-10 15:37 - step6-security-groups-p1-vpc-list.png - Created

Security groups in p1-vpc: p1-sg-alb sg-00e39906c1d91de7e, p1-sg-app sg-0dc782d147f0112b, p1-sg-db sg-063e10e3231772a8b. Banner confirms p1-sg-db created.



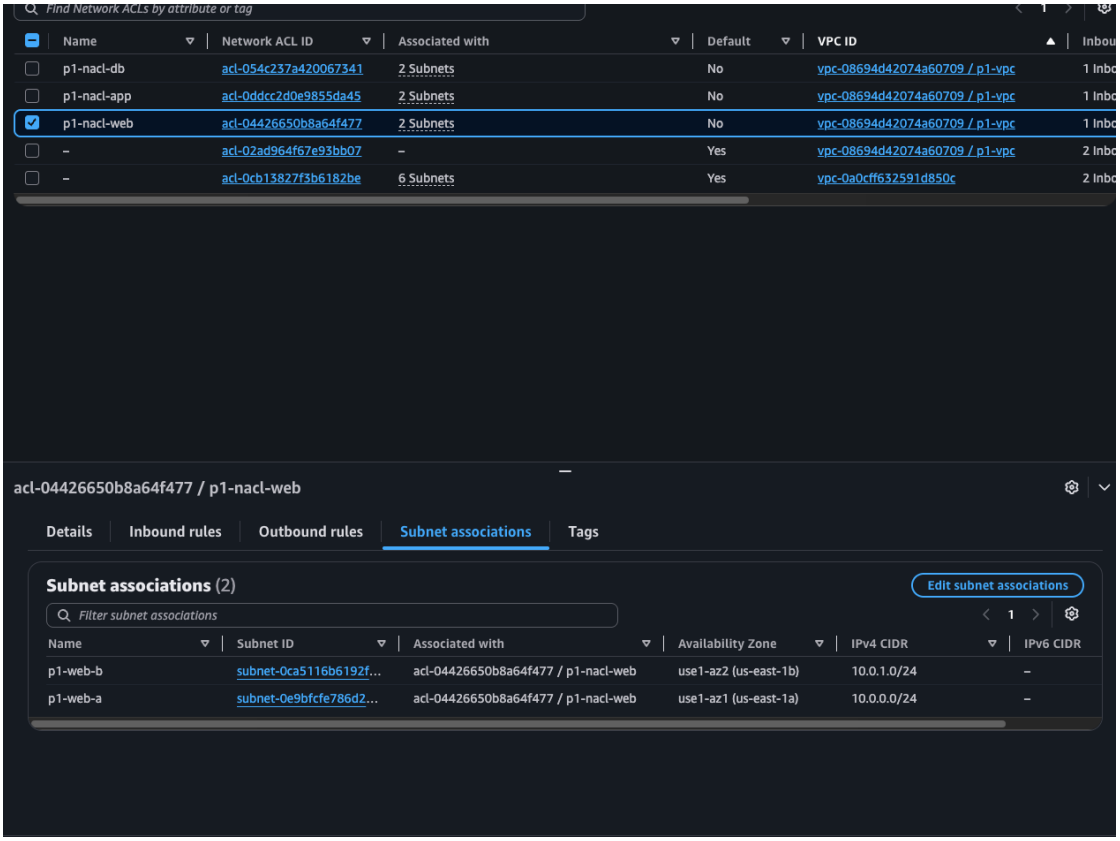
File: screenshots/security-groups/step6-security-groups-p1-vpc-list.png

E15 - 2026-11-10 15:39 - (paste, no screenshot file) - Step 6 complete

Inbound rules, 1 per group, pasted: p1-sg-alb HTTP 80 from 0.0.0.0/0; p1-sg-app HTTP 80 from p1-sg-alb; p1-sg-db MySQL 3306 from p1-sg-app. Chain verified.

E16 - 2026-11-10 15:47 - step7-nacl-p1-nacl-web-2-web-subnets.png - Associations verified

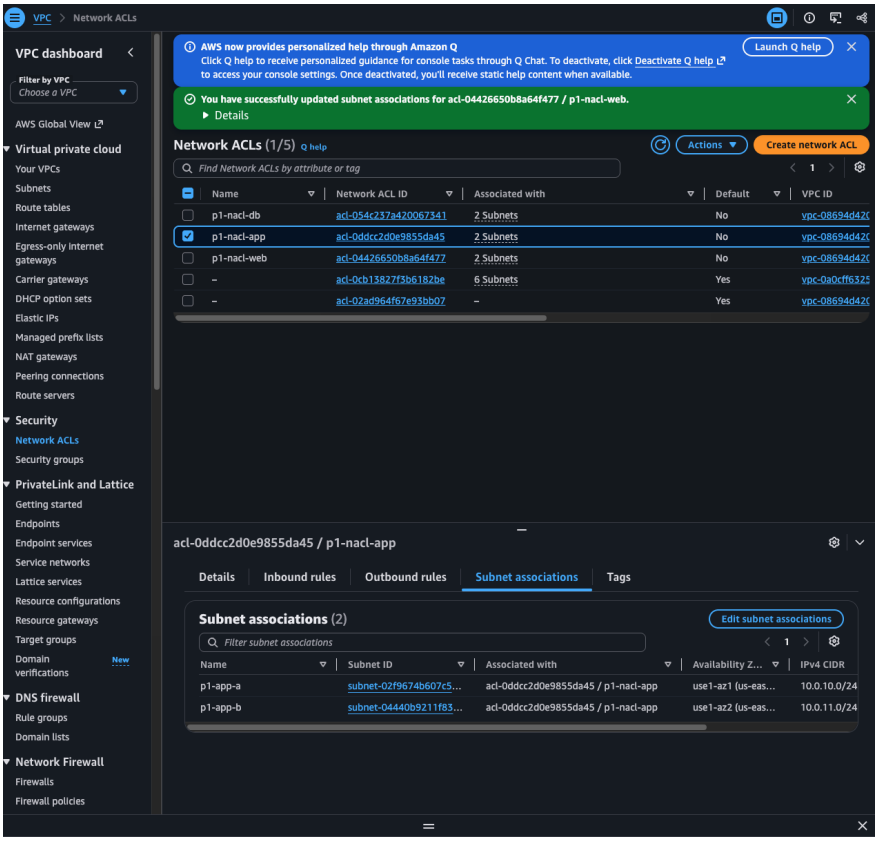
p1-nacl-web acl-04426650b8a64f477, 2 subnets: p1-web-a, p1-web-b.



File: screenshots/security-groups/step7-nacl-p1-nacl-web-2-web-subnets.png

E17 - 2026-11-10 15:48 - step7-nacl-p1-nacl-app-2-app-subnets.png - Associations verified

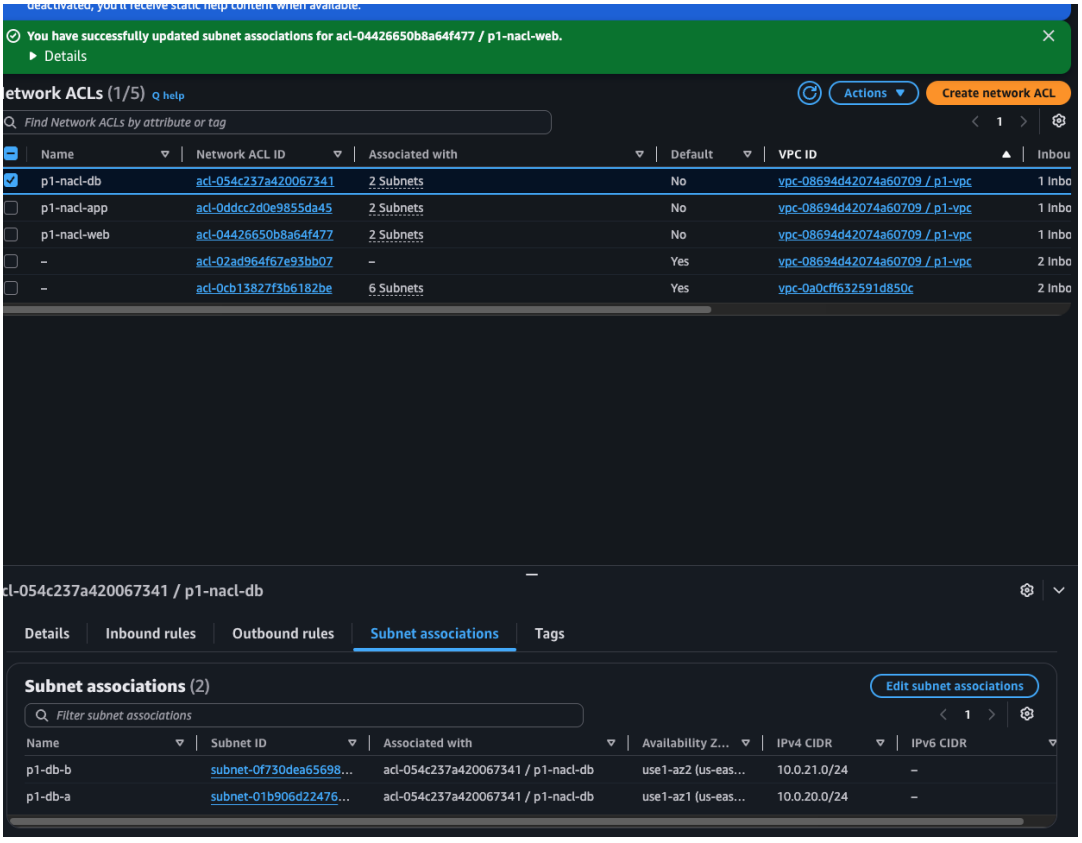
p1-nacl-app acl-0ddcc2d0e9855da45, 2 subnets: p1-app-a, p1-app-b.



File: screenshots/security-groups/step7-nacl-p1-nacl-app-2-app-subnets.png

E18 - 2026-11-10 15:48 - step7-nacl-p1-nacl-db-2-db-subnets.png - Associations verified

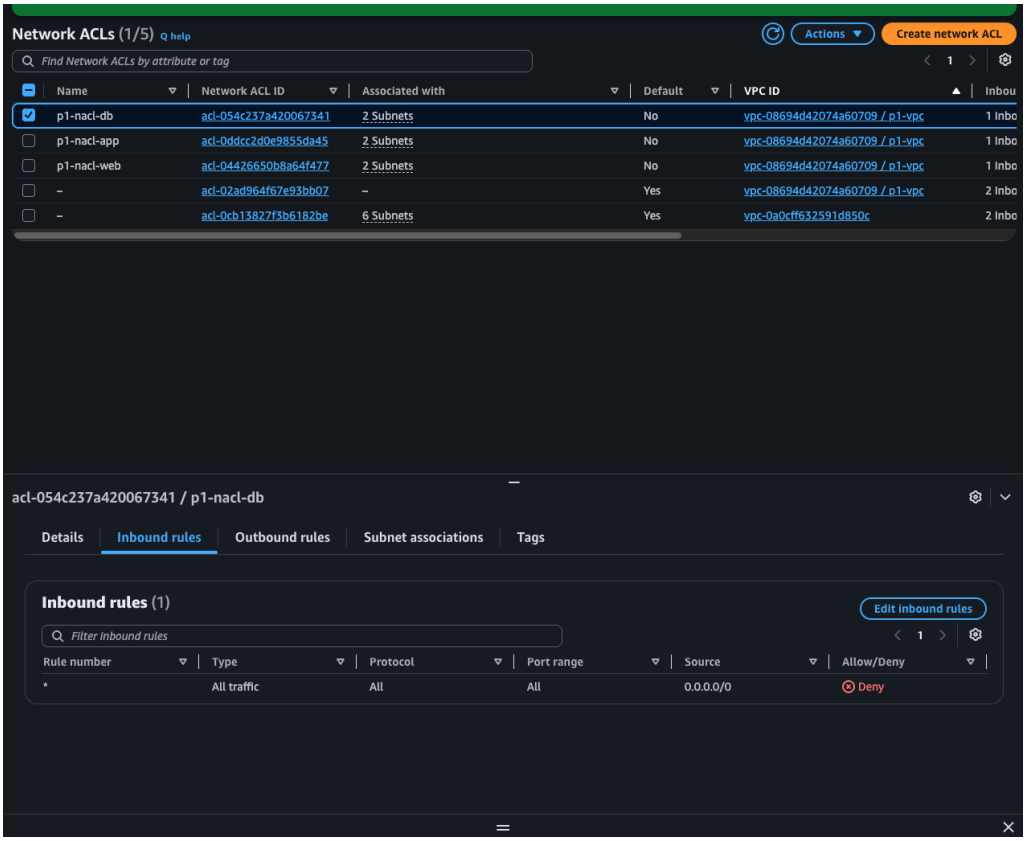
p1-nacl-db acl-054c237a420067341, 2 subnets: p1-db-a, p1-db-b.



File: screenshots/security-groups/step7-nacl-p1-nacl-db-2-db-subnets.png

E19 - 2026-11-10 15:50 - step7-nacl-p1-nacl-db-inbound-only-deny.png - Explained, not an error

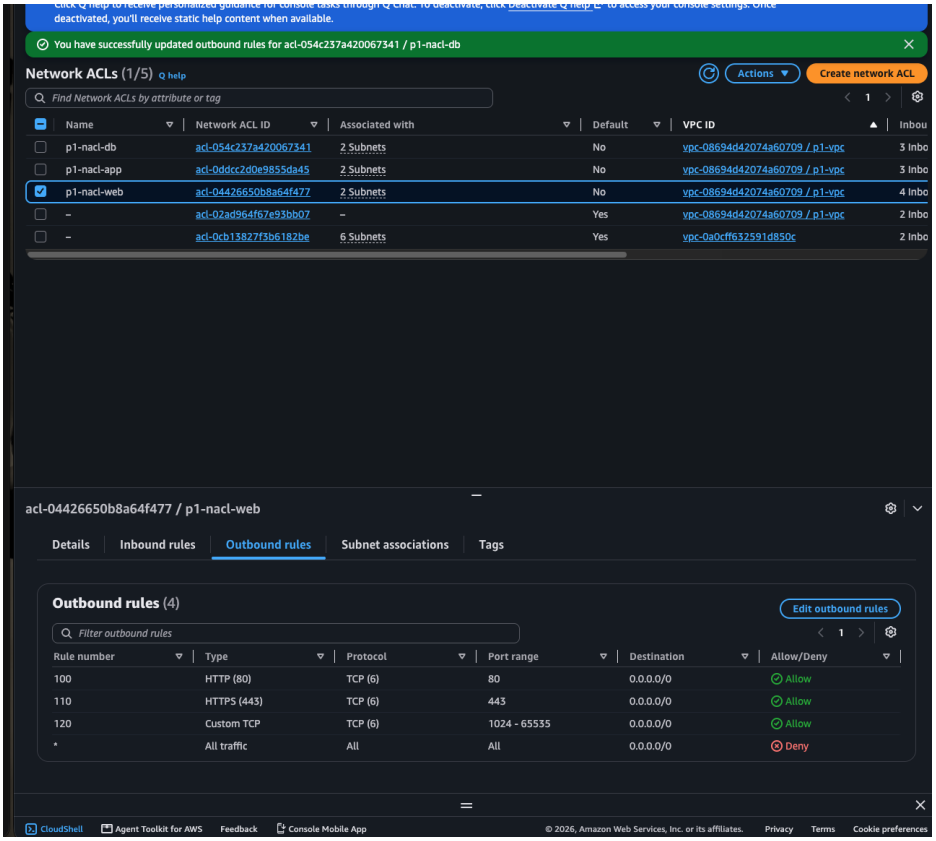
p1-nacl-db inbound rules (1): only * Deny. No allow rules yet. A new custom NACL starts with only * Deny, that is normal.



File: screenshots/security-groups/step7-nacl-p1-nacl-db-inbound-only-deny.png

E20 - 2026-11-10 16:05 - step7-nacl-p1-nacl-web-outbound-correct.png - Correct

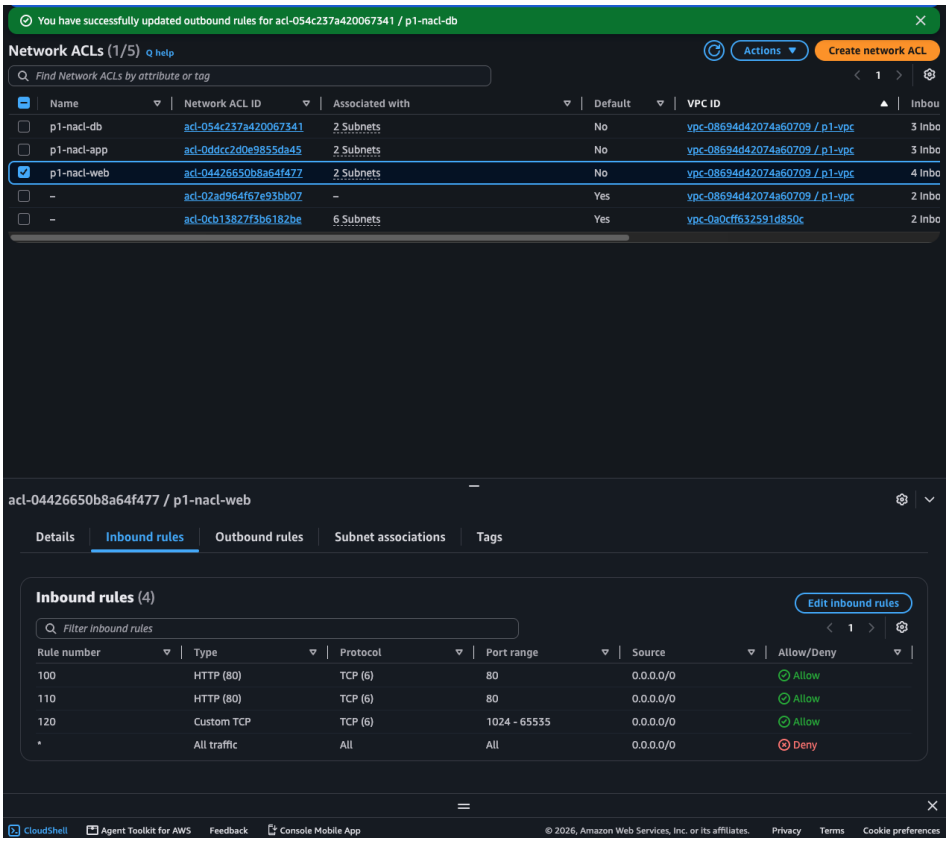
p1-nacl-web outbound (4): 100 HTTP 80, 110 HTTPS 443, 120 TCP 1024-65535, all to 0.0.0.0/0 Allow; * Deny.



File: screenshots/security-groups/step7-nacl-p1-nacl-web-outbound-correct.png

E21 - 2026-11-10 16:05 - step7-nacl-p1-nacl-web-inbound-110-wrong-http.png - One rule wrong

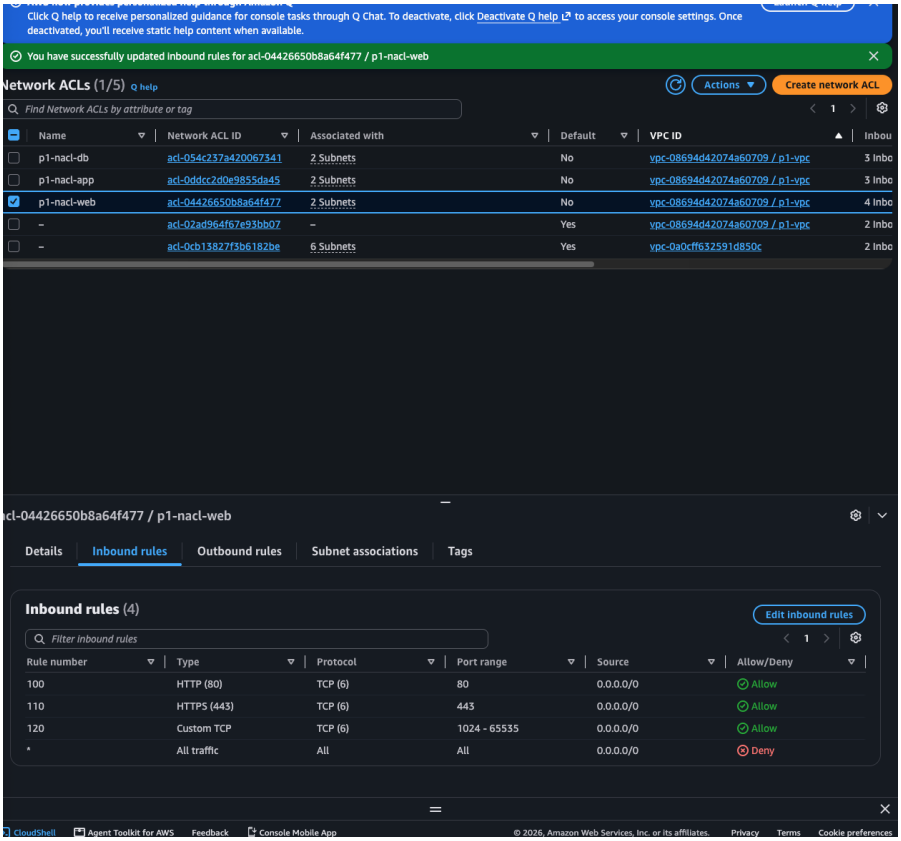
p1-nacl-web inbound (4): 100 HTTP 80 Allow, 110 HTTP 80 again (should be HTTPS 443), 120 TCP 1024-65535 Allow; * Deny.



File: screenshots/security-groups/step7-nacl-p1-nacl-web-inbound-110-wrong-http.png

E22 - 2026-11-10 20:19 - step7-nacl-p1-nacl-web-inbound-110-fixed-https.png - Fixed

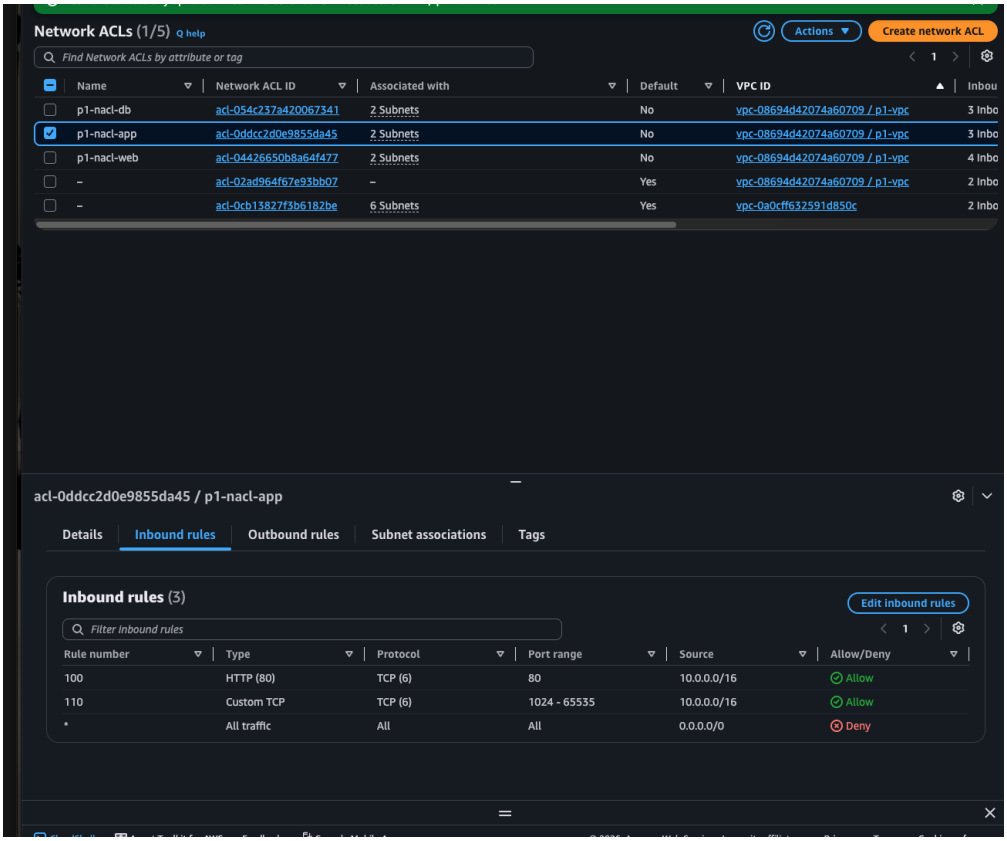
p1-nacl-web inbound (4) now correct: 100 HTTP 80, 110 HTTPS 443, 120 TCP 1024-65535, all from 0.0.0.0/0 Allow; * Deny.



File: screenshots/security-groups/step7-nacl-p1-nacl-web-inbound-110-fixed-https.png

E23 - 2026-11-10 20:21 - step7-nacl-p1-nacl-app-inbound-correct.png - Correct as designed

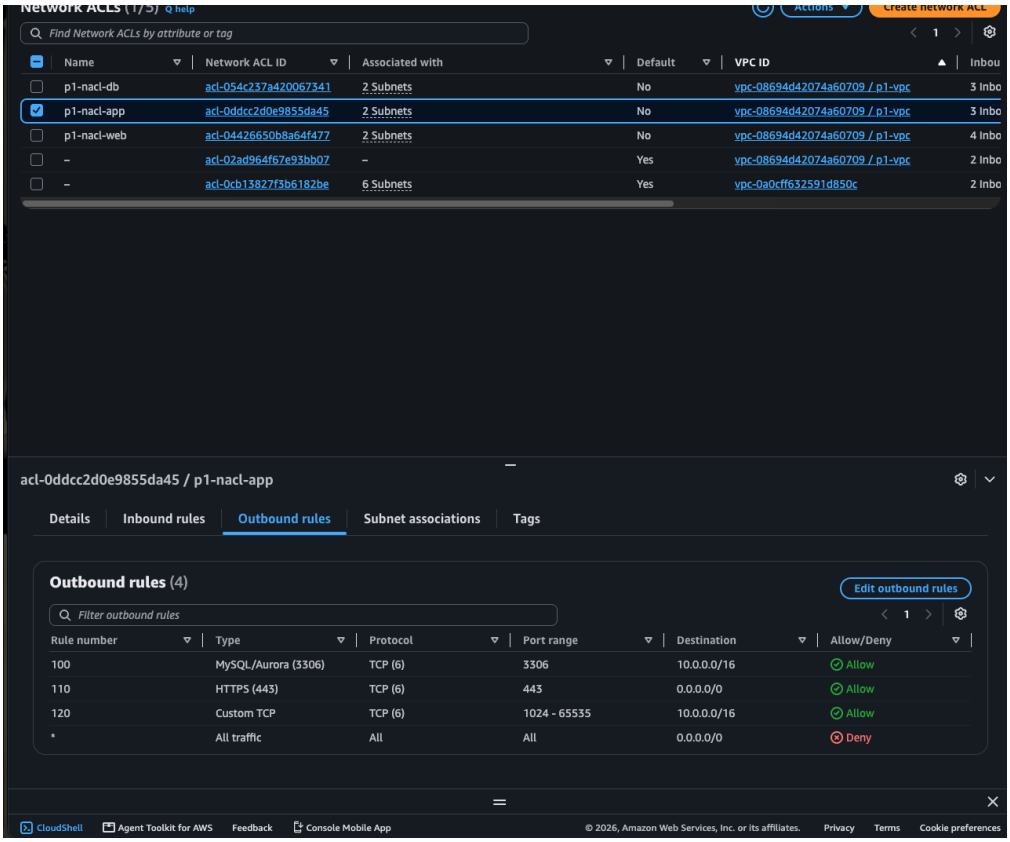
p1-nacl-app inbound (3): 100 HTTP 80 from 10.0.0.0/16, 110 TCP 1024-65535 from 10.0.0.0/16, * Deny. Original Step 7 version; later amended in Step 9.



File: screenshots/security-groups/step7-nacl-p1-nacl-app-inbound-correct.png

E24 - 2026-11-10 20:21 - step7-nacl-p1-nacl-app-outbound-correct.png - Correct as designed

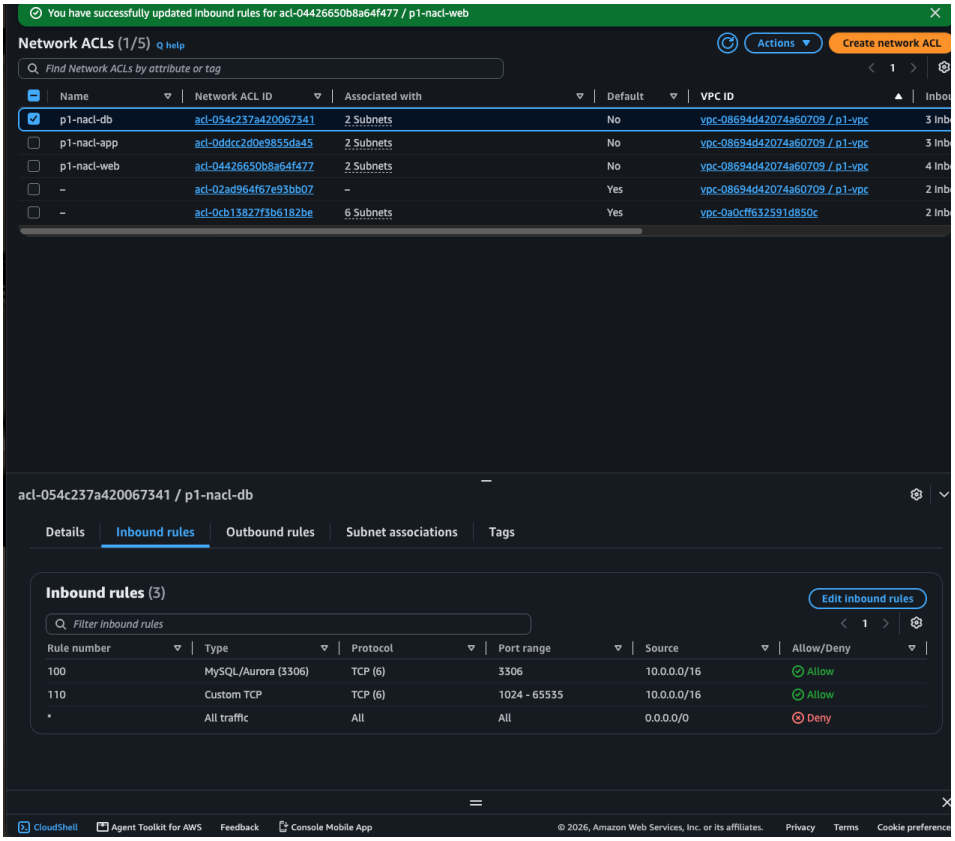
p1-nacl-app outbound (4): 100 MySQL 3306 to 10.0.0.0/16, 110 HTTPS 443 to 0.0.0.0/0, 120 TCP 1024-65535 to 10.0.0.0/16, * Deny. Original version; DNS rule 130 added in Step 9.



File: screenshots/security-groups/step7-nacl-p1-nacl-app-outbound-correct.png

E25 - 2026-11-10 20:21 - step7-nacl-p1-nacl-db-inbound-correct.png - Correct

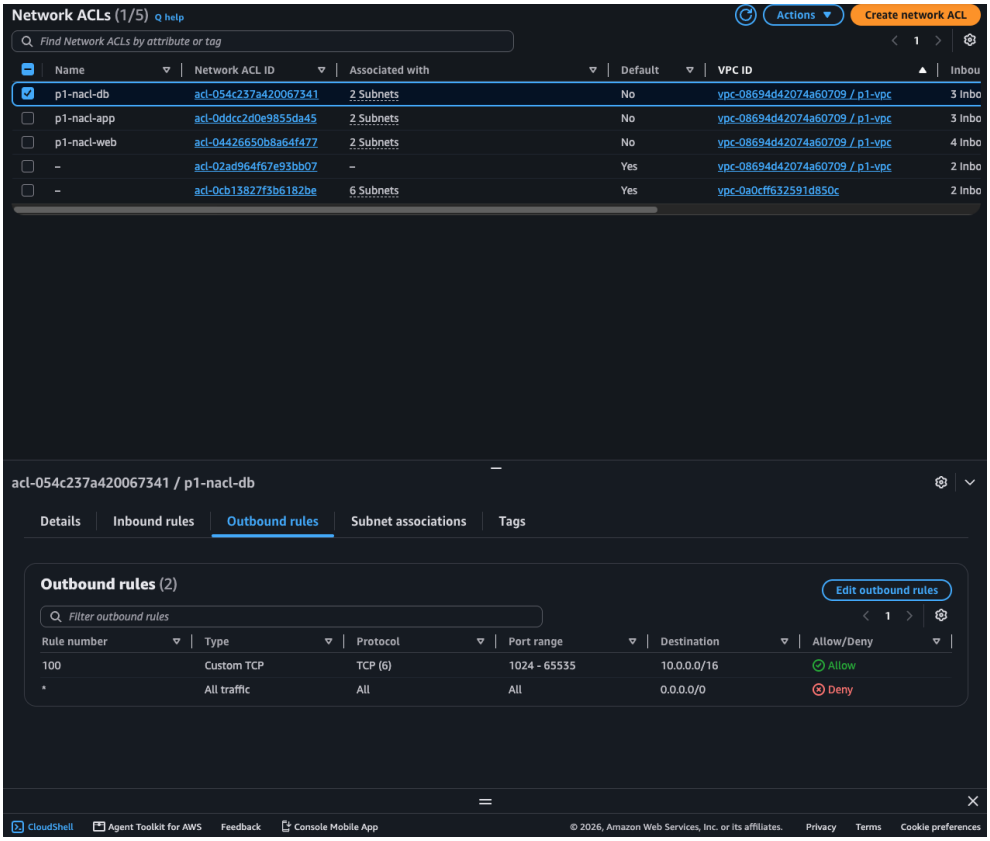
p1-nacl-db inbound (3): 100 MySQL 3306 from 10.0.0.0/16, 110 TCP 1024-65535 from 10.0.0.0/16, * Deny.



File: screenshots/security-groups/step7-nacl-p1-nacl-db-inbound-correct.png

E26 - 2026-11-10 20:21 - step7-nacl-p1-nacl-db-outbound-correct.png - Correct - Step 7 complete

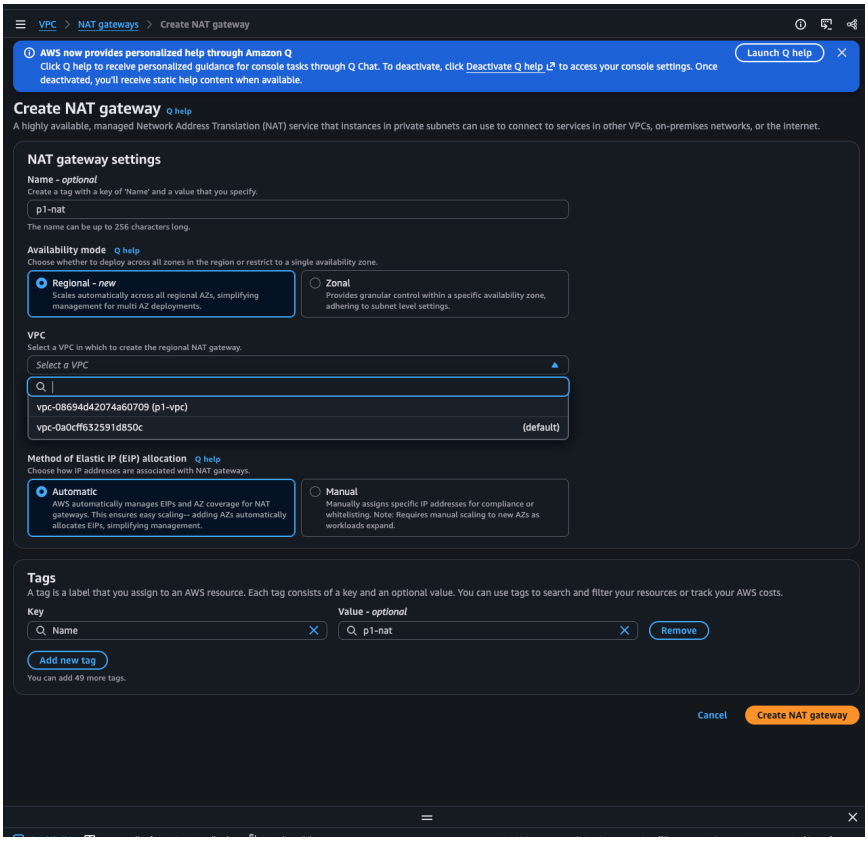
p1-nacl-db outbound (2): 100 TCP 1024-65535 to 10.0.0.0/16, * Deny. No 0.0.0.0/0 allow rule anywhere.



File: screenshots/security-groups/step7-nacl-p1-nacl-db-outbound-correct.png

E27 - 2026-11-10 20:24 - step8-create-nat-gateway-regional-no-subnet.png - Form trap noted

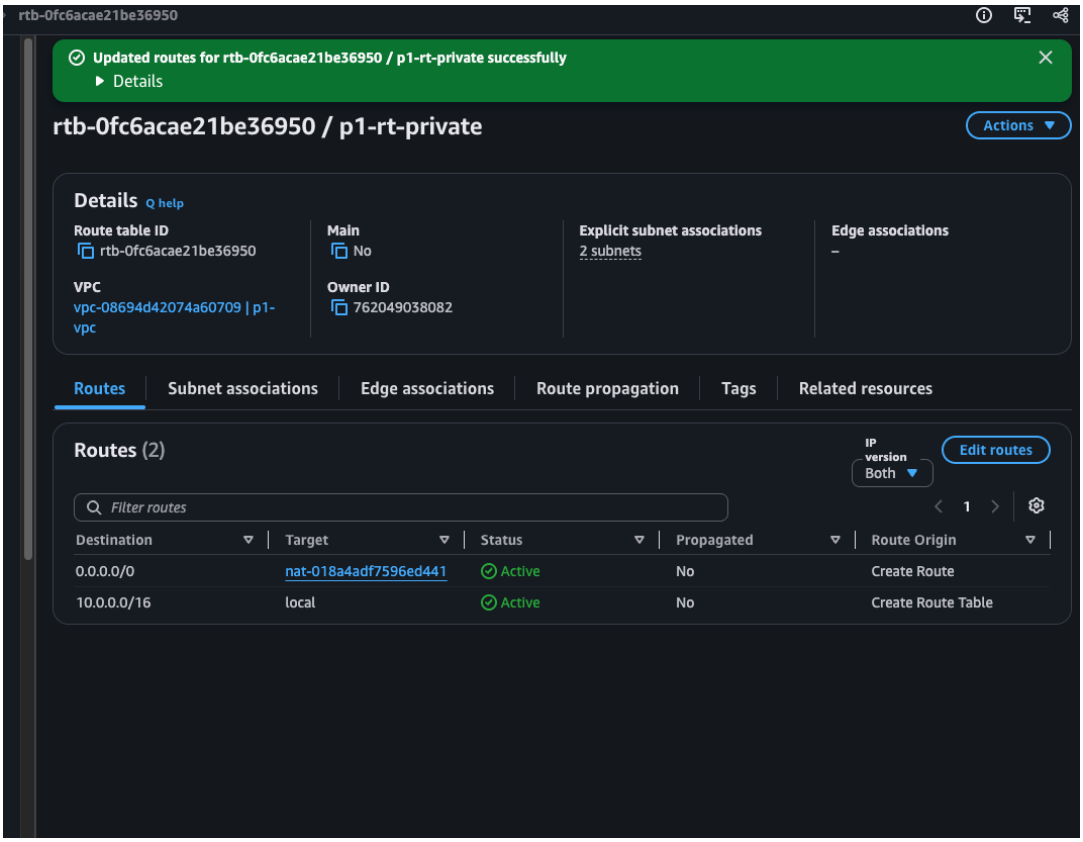
Create NAT gateway form, Availability mode Regional, which hides the Subnet field. Switch to Zonal to pick p1-web-a.



File: screenshots/vpc-subnets/step8-create-nat-gateway-regional-no-subnet.png

E28 - 2026-11-10 20:29 - step8-route-table-p1-rt-private-nat-route.png - NAT route verified

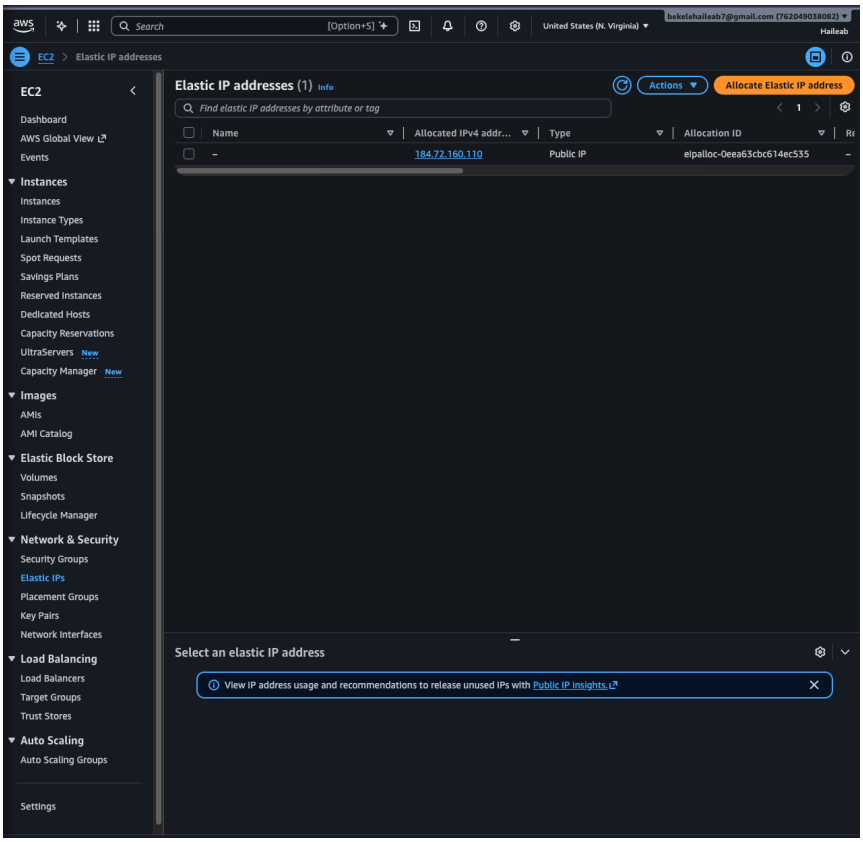
p1-rt-private routes (2): 0.0.0.0/0 to nat-018a4adf7596ed441 Active, 10.0.0.0/16 local Active. Original p1-nat ID nat-018a4adf7596ed441.



File: screenshots/vpc-subnets/step8-route-table-p1-rt-private-nat-route.png

E29 - 2026-11-10 20:32 - step8-elastic-ip-184-72-160-110.png - EIP found

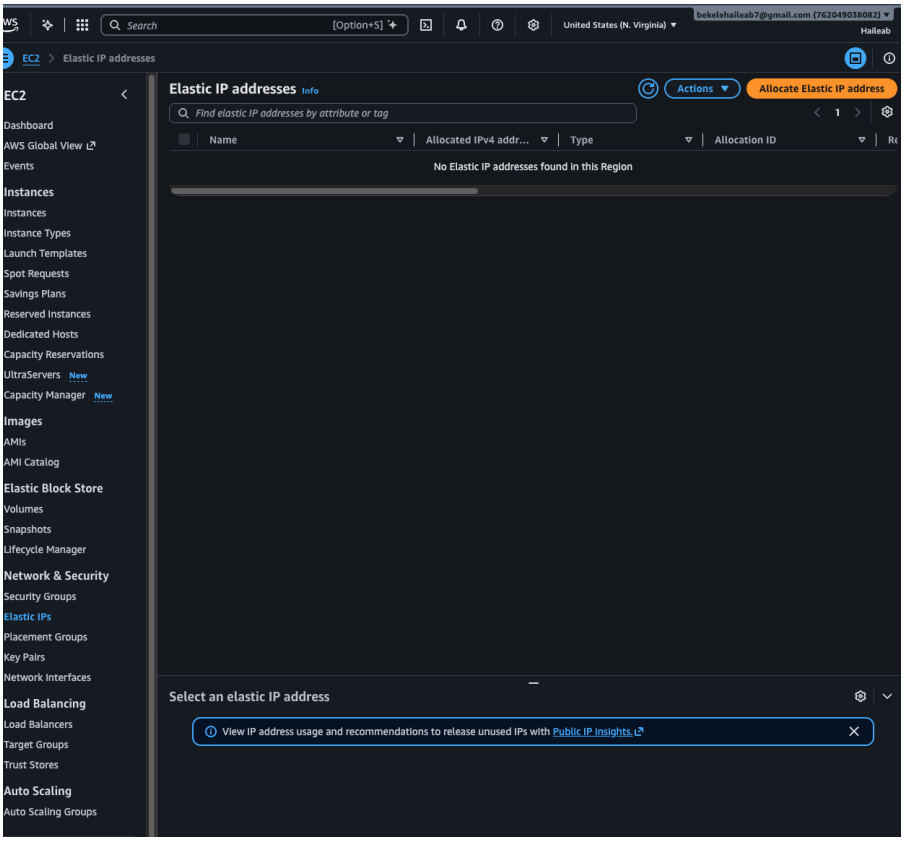
Elastic IP addresses (1): 184.72.160.110, allocation eipalloc-0eea63cbc614ec535, the p1-nat EIP, not yet released.



File: screenshots/vpc-subnets/step8-elastic-ip-184-72-160-110.png

E30 - 2026-11-10 20:33 - step8-elastic-ips-empty.png - EIP released

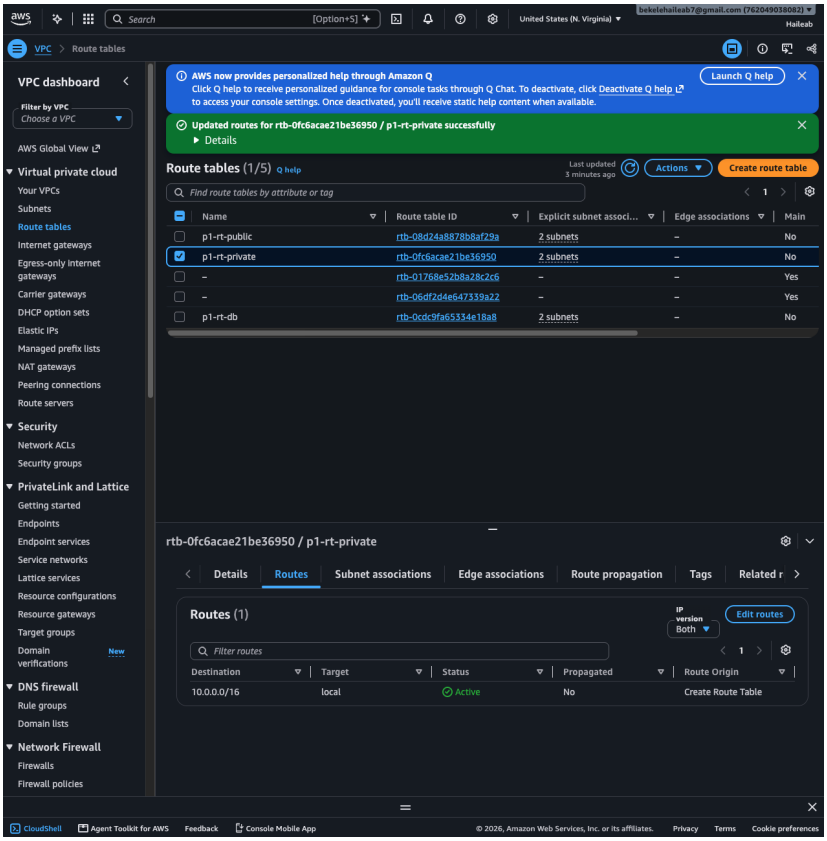
Elastic IP addresses: No Elastic IP addresses found in this Region. EIP released.



File: screenshots/vpc-subnets/step8-elastic-ips-empty.png

E31 - 2026-11-10 20:34 - step8-route-table-p1-rt-private-local-only.png - Step 8 complete

p1-rt-private routes (1): 10.0.0.0/16 local Active only, no 0.0.0.0/0 route. NAT deleted, route removed, EIP released in one session.



File: screenshots/vpc-subnets/step8-route-table-p1-rt-private-local-only.png

E32 - 2026-11-10 20:47 - step9-asg-load-balancing-p1-tg-app.png - As expected

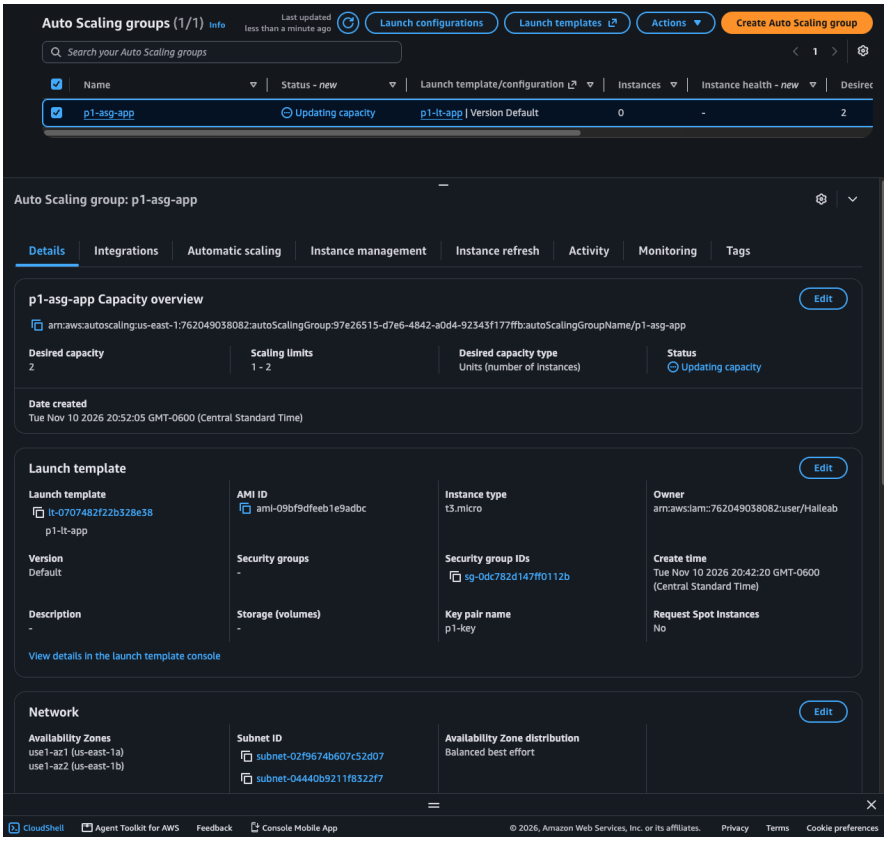
ASG load balancing page: attach to existing target group p1-tg-app HTTP selected, load balancer not associated yet, expected before Part D.



File: screenshots/load-balancer/step9-asg-load-balancing-p1-tg-app.png

E33 - 2026-11-10 20:52 - step9-asg-p1-asg-app-created.png - ASG created

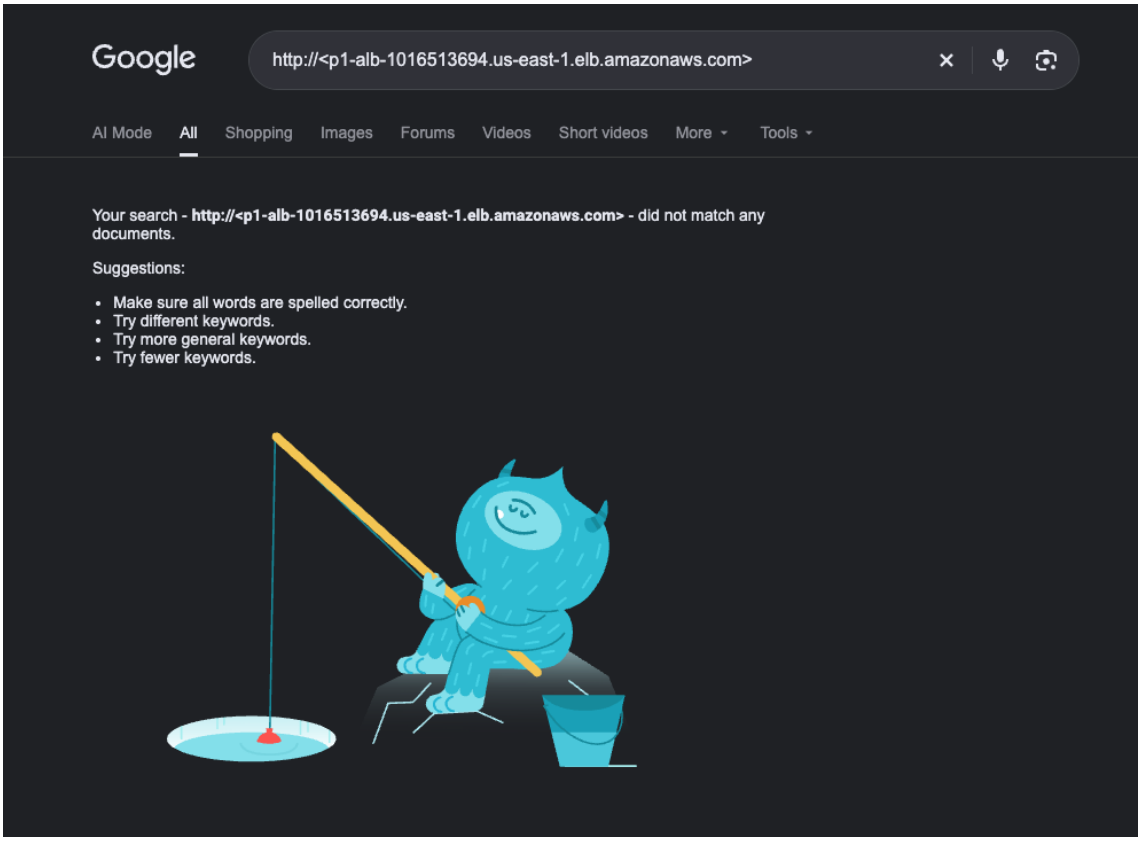
p1-asg-app created, Desired 2, Min 1, Max 2, Instances 0 launching. Launch template p1-lt-app lt-0707482f22b328e38, AMI ami-09bf9dfeeb1e9adbc, t3.micro, p1-sg-app, key p1-key, subnets p1-app-a/b.



File: screenshots/load-balancer/step9-asg-p1-asg-app-created.png

E34 - 2026-11-10 21:05 - step9-alb-browser-test-google-search-wrong-url.png - Test failed - URL entry only

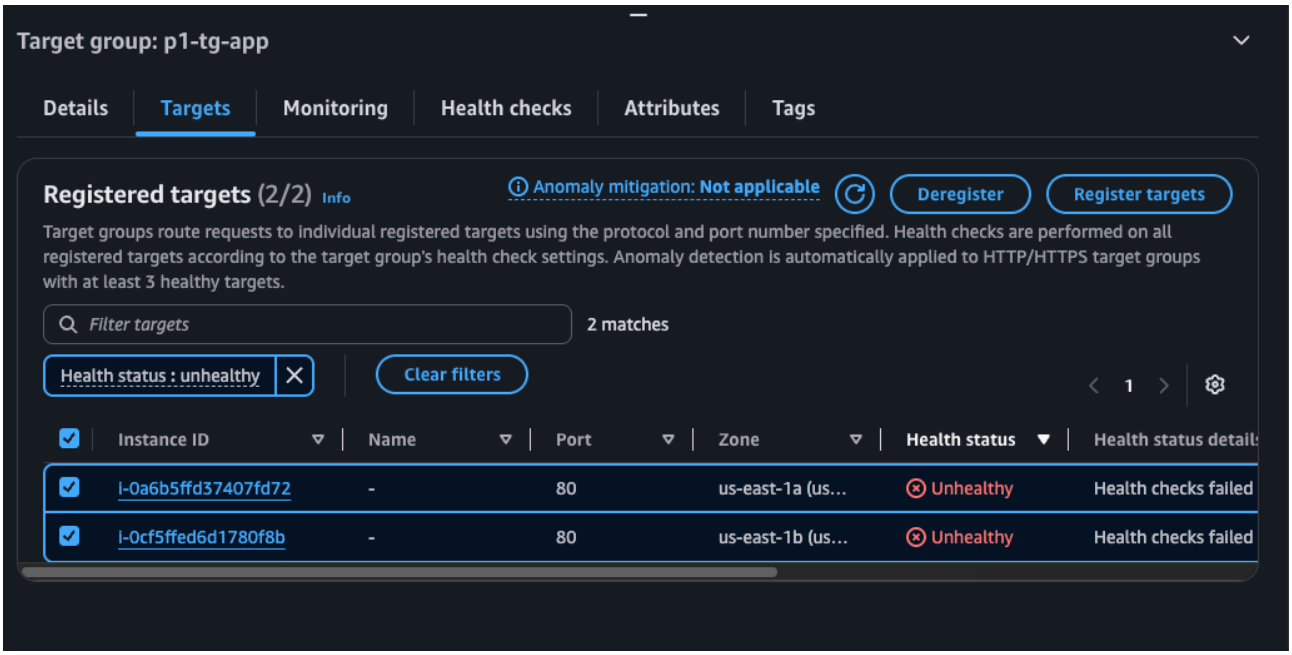
Google search page, no results. ALB DNS identified: p1-alb-1016513694.us-east-1.elb.amazonaws.com. URL was typed with angle brackets into search, not the address bar.



File: screenshots/load-balancer/step9-alb-browser-test-google-search-wrong-url.png

E35 - 2026-11-10 21:07 - step9-target-group-2-unhealthy.png - Unhealthy - NAT missing

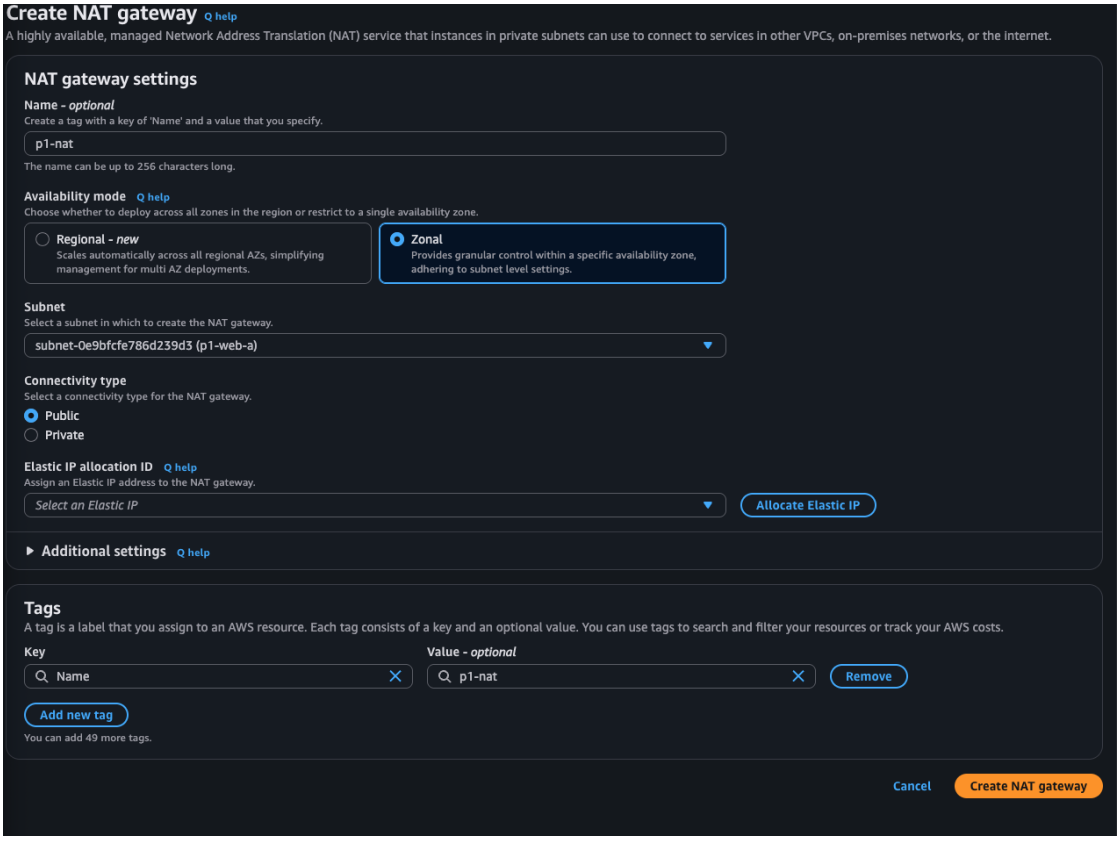
p1-tg-app registered targets (2/2), both Unhealthy, Health checks failed: i-0a6b5ffd37407fd72 (1a), i-0cf5ffed6d1780f8b (1b). No healthy targets, ALB has nothing to forward to.



File: screenshots/load-balancer/step9-target-group-2-unhealthy.png

E36 - 2026-11-10 21:11 - step9-recreate-nat-zonal-p1-web-a.png - NAT recreation

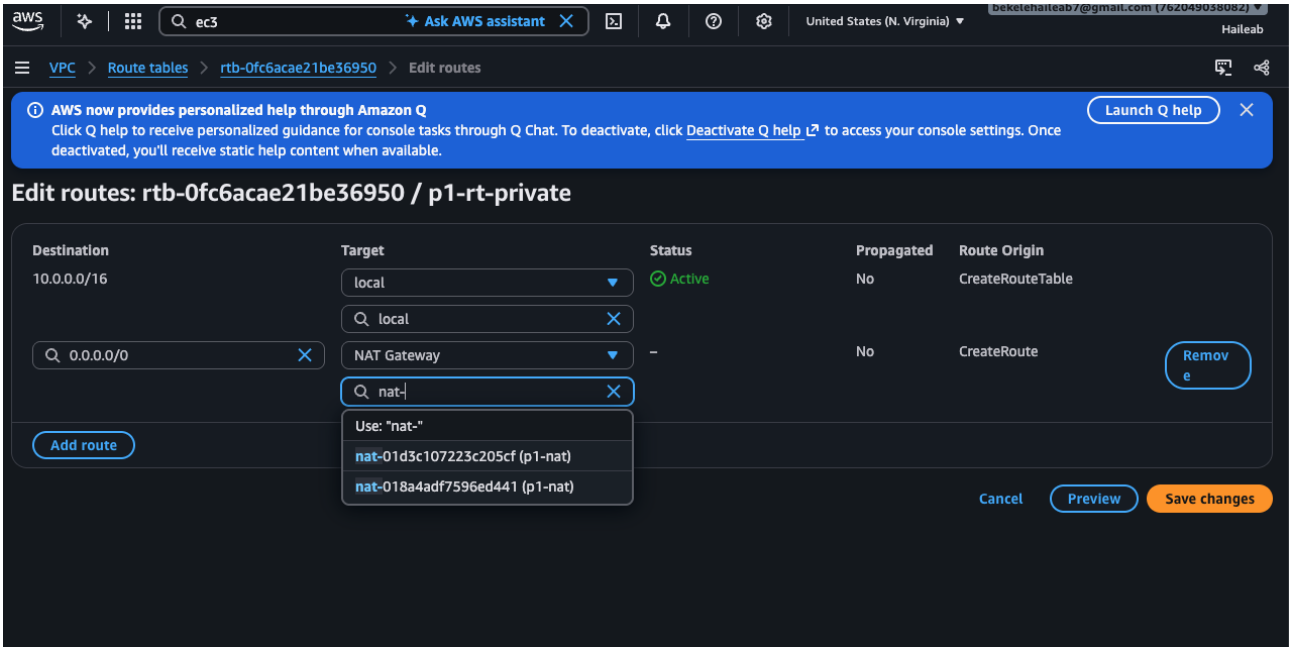
Recreate NAT form, Zonal mode, subnet p1-web-a selected. In Zonal mode there is no VPC field, the subnet decides the VPC.



File: screenshots/vpc-subnets/step9-recreate-nat-zonal-p1-web-a.png

E37 - 2026-11-10 21:14 - step9-nat-route-choose-new-nat.png - Route target chosen

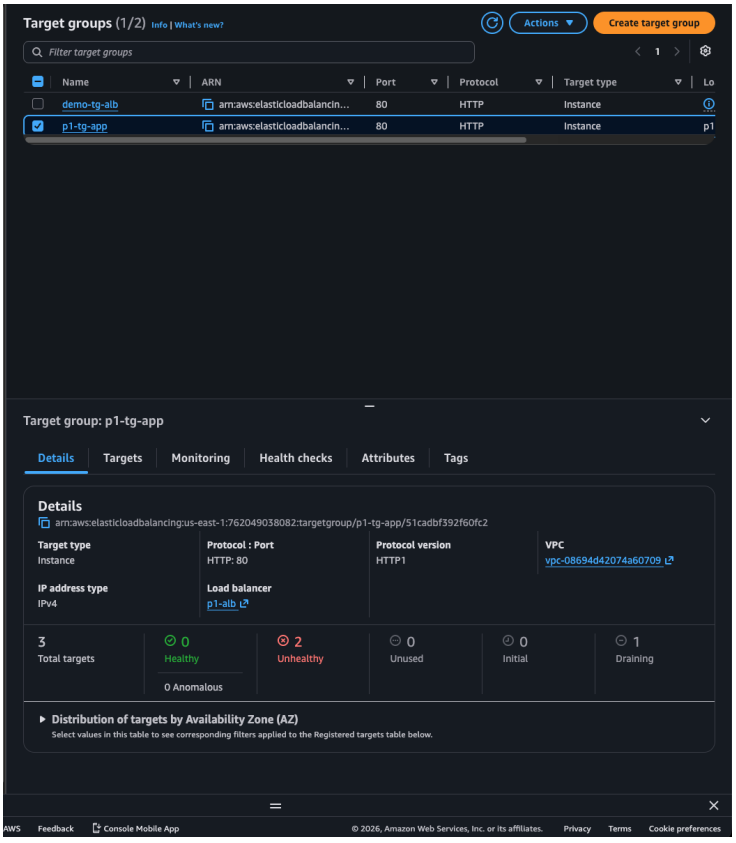
Edit routes for p1-rt-private, 0.0.0.0/0, target NAT Gateway, dropdown shows new nat-01d3c107223c205cf and old deleted nat-018a4adf7596ed441. Pick the new one.



File: screenshots/vpc-subnets/step9-nat-route-choose-new-nat.png

E38 - 2026-11-10 21:20 - step9-target-group-3-total-2-unhealthy-1-draining.png - In transition

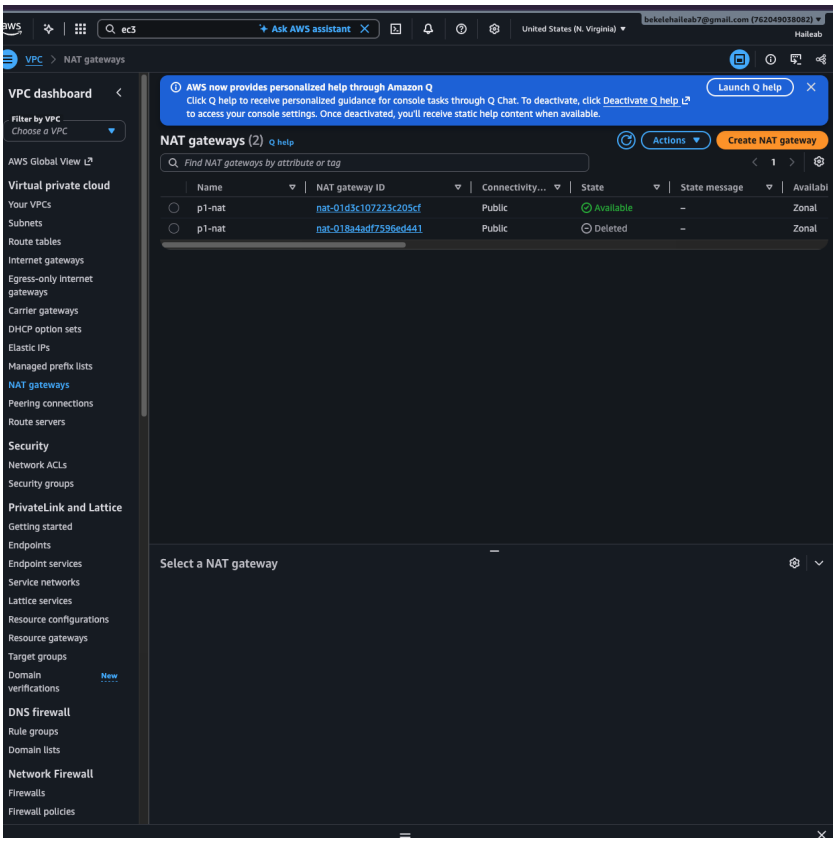
p1-tg-app Total 3, Healthy 0, Unhealthy 2, Draining 1. The Draining target is an old terminated instance.



File: screenshots/load-balancer/step9-target-group-3-total-2-unhealthy-1-draining.png

E39 - 2026-11-10 21:20 - step9-nat-gateways-new-available-old-deleted.png - New NAT up

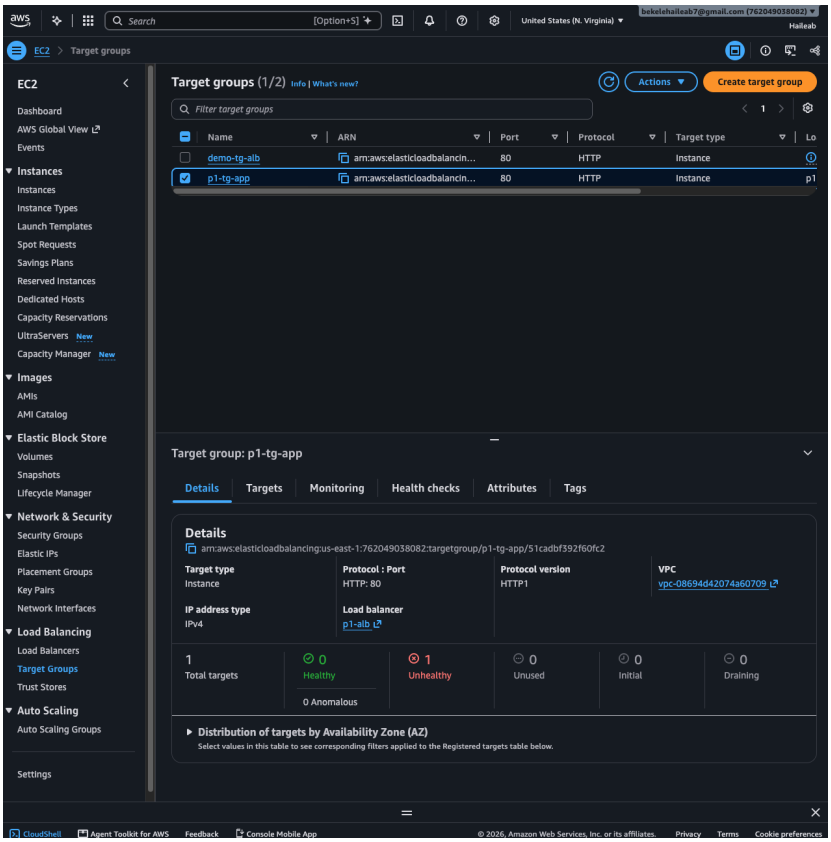
NAT gateways (2): nat-01d3c107223c205cf Public Available Zonal (new), nat-018a4adf7596ed441 Public Deleted Zonal (old).



File: screenshots/vpc-subnets/step9-nat-gateways-new-available-old-deleted.png

E40 - 2026-11-10 21:38 - step9-target-group-1-total-1-unhealthy-after-dns-fix.png - In transition

p1-tg-app Total 1, Healthy 0, Unhealthy 1. Only 1 of 2 replacements had registered yet after the DNS fix.



File: screenshots/load-balancer/step9-target-group-1-total-1-unhealthy-after-dns-fix.png

E41 - 2026-11-10 21:54 - (paste, no screenshot file) - DNS fix alone did not solve it

Targets after DNS fix, still Unhealthy: i-00e10a16bb96c1321 (1a, launch 21:39), i-06c5699107e9dbc39 (1b, launch 21:37), Health checks failed, 15-17 min after launch.

E42 - 2026-11-10 21:56 - (paste, no screenshot file) - NACL DNS verified

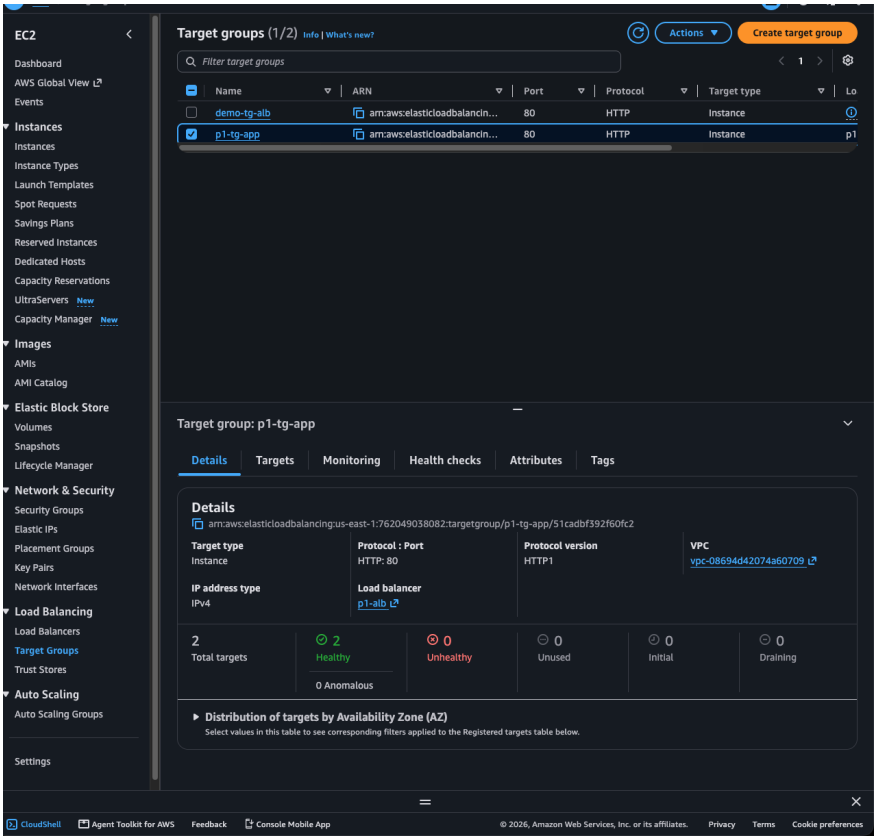
p1-nacl-app rules verified: inbound 100 HTTP 80, 110 TCP ephemeral, 120 UDP ephemeral, * Deny; outbound 100 MySQL 3306, 110 HTTPS 443, 120 TCP ephemeral, 130 DNS UDP 53, * Deny. Both DNS rules present and Allow.

E43 - 2026-11-10 21:57 - (paste, no screenshot file) - Root cause found

Instance system log, i-00e10a16bb96c1321, 10.0.10.122: dnf Curl error (28) Timeout to S3 al2023 repos, No match for httpd, Failed to start httpd.service, /var/www/html/index.html No such file. A timeout, not a DNS failure. Inbound NACL ephemeral TCP only from 10.0.0.0/16 denied return traffic from public S3 IPs.

E44 - 2026-11-10 22:07 - step9-target-group-2-healthy.png - Healthy - fix confirmed

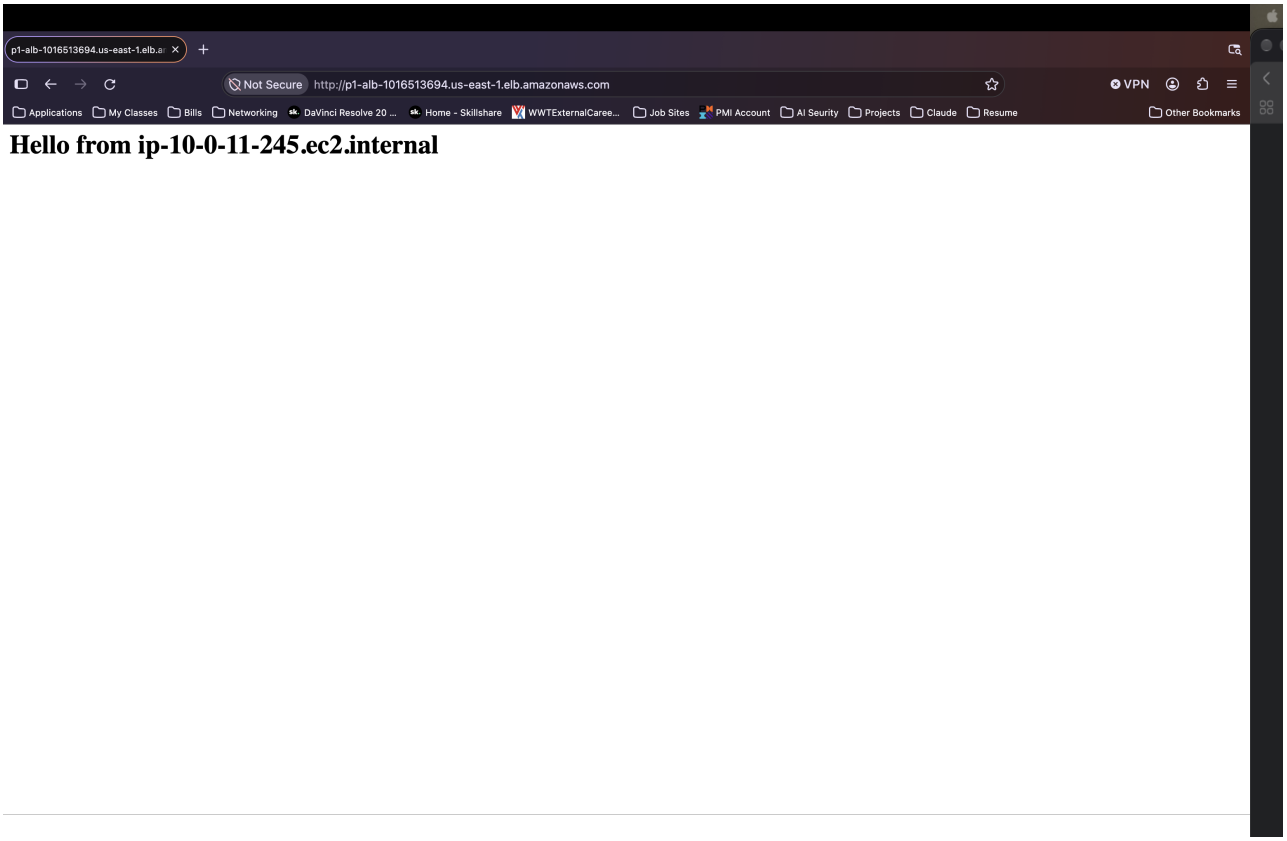
p1-tg-app Total 2, Healthy 2, Unhealthy 0. Fix: p1-nacl-app inbound rule 130, Custom TCP 1024-65535 from 0.0.0.0/0 Allow, then terminate both instances again.



File: screenshots/load-balancer/step9-target-group-2-healthy.png

E45 - 2026-11-10 22:10 - step9-alb-browser-test-hello-from-ip-10-0-11-245.png - Step 9 complete - ALB works

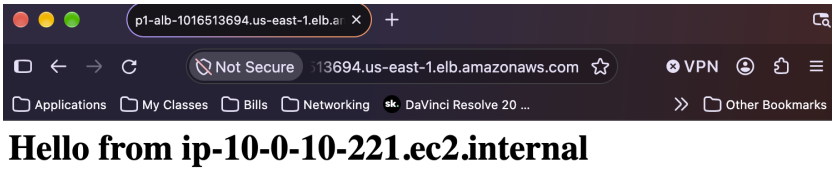
Browser, http://p1-alb-1016513694.us-east-1.elb.amazonaws.com, page: Hello from ip-10-0-11-245.ec2.internal, app-b server, us-east-1b.



File: screenshots/load-balancer/step9-alb-browser-test-hello-from-ip-10-0-11-245.png

E46 - 2026-11-10 22:11 - step9-alb-browser-test-hello-from-ip-10-0-10-221.png - Both AZs serve traffic

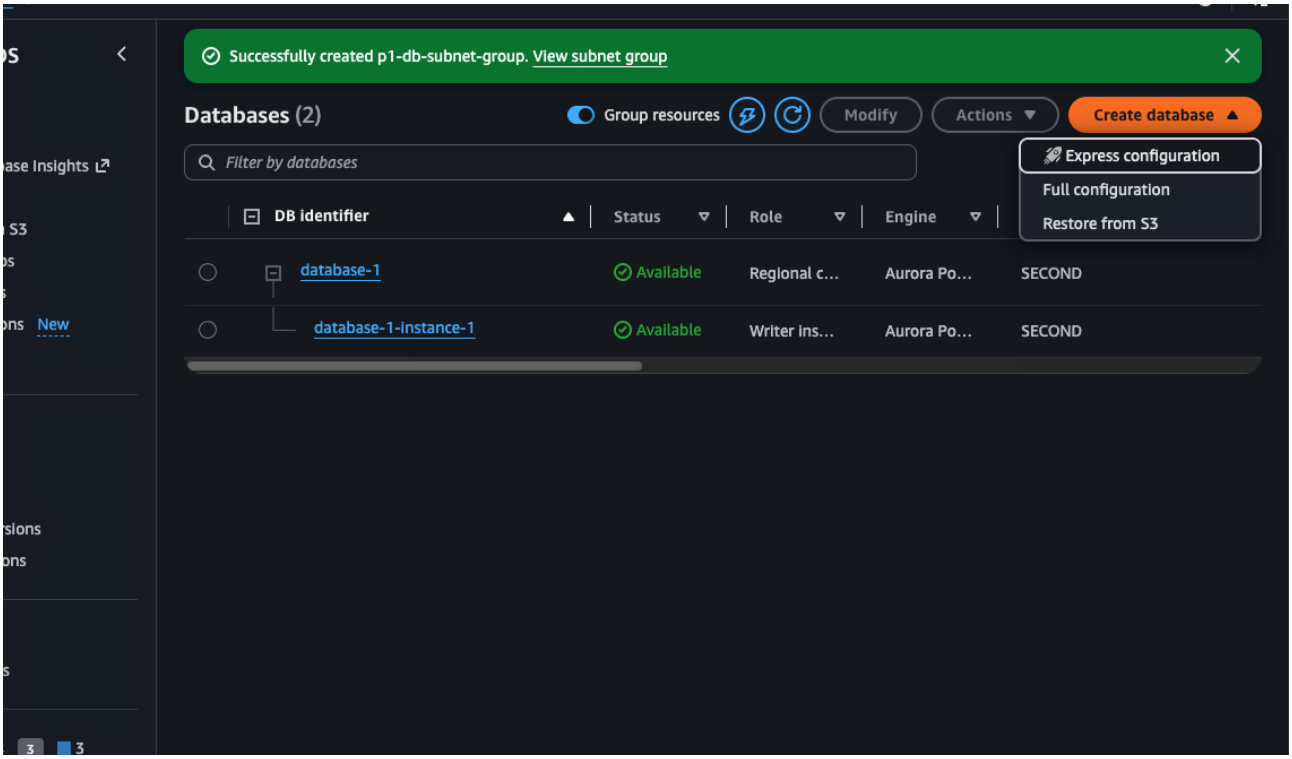
Browser, after refresh: Hello from ip-10-0-10-221.ec2.internal, app-a server, us-east-1a. Load balancing across both AZs proven.



File: screenshots/load-balancer/step9-alb-browser-test-hello-from-ip-10-0-10-221.png

E47 - 2026-11-10 22:16 - step10-db-subnet-group-created.png - Subnet group created

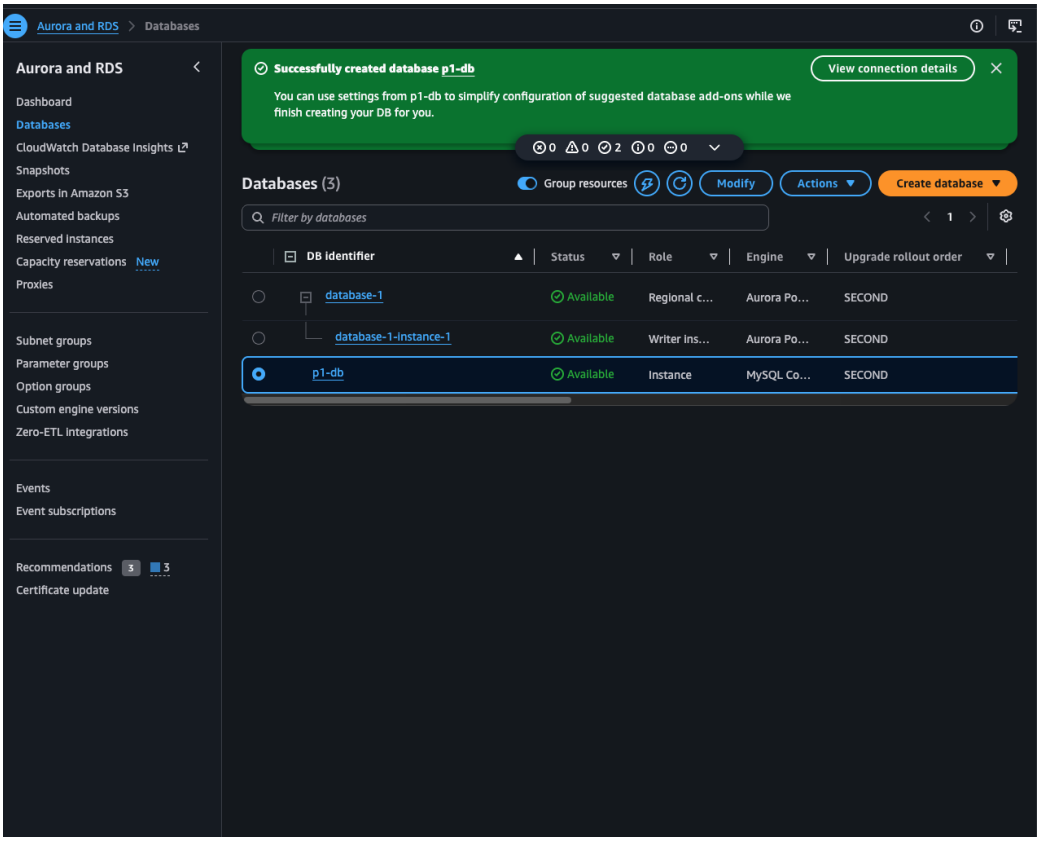
Banner: Successfully created p1-db-subnet-group. Databases (2) already exist: database-1 and database-1-instance-1, Aurora PostgreSQL, pre-existing, do not touch.



File: screenshots/database/step10-db-subnet-group-created.png

E48 - 2026-11-10 22:28 - step10-rds-p1-db-available.png - Part A complete

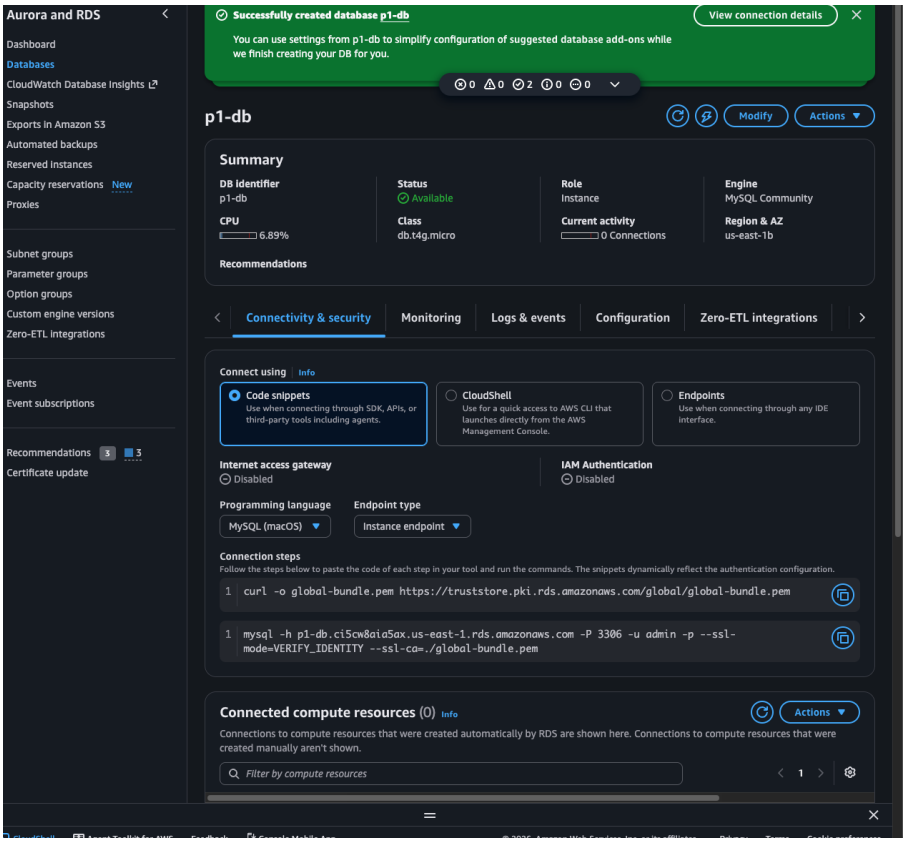
Banner: Successfully created database p1-db. Databases (3): p1-db Available, Role Instance, Engine MySQL Community.



File: screenshots/database/step10-rds-p1-db-available.png

E49 - 2026-11-10 22:30 - step10-rds-p1-db-endpoint.png - Endpoint verified

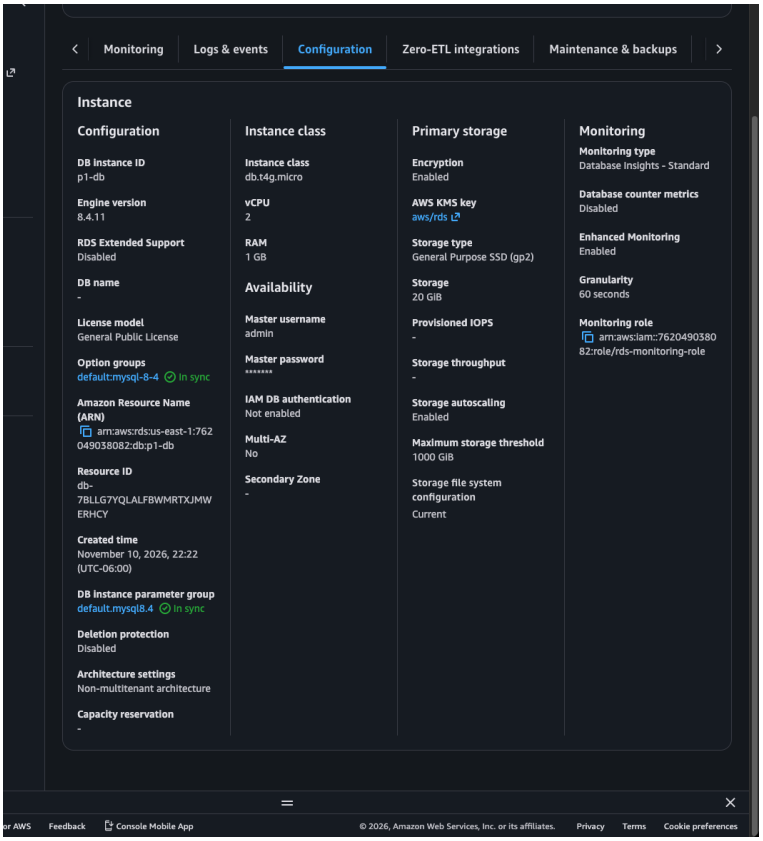
p1-db details: Available, MySQL Community, CPU 6.89%, 0 Connections, Class db.t4g.micro (not db.t3.micro), AZ us-east-1b, Endpoint p1-db.ci5cw8aia5ax.us-east-1.rds.amazonaws.com, Port 3306.



File: screenshots/database/step10-rds-p1-db-endpoint.png

E50 - 2026-11-10 22:30 - step10-rds-p1-db-configuration.png - Configuration verified, 3 corrections

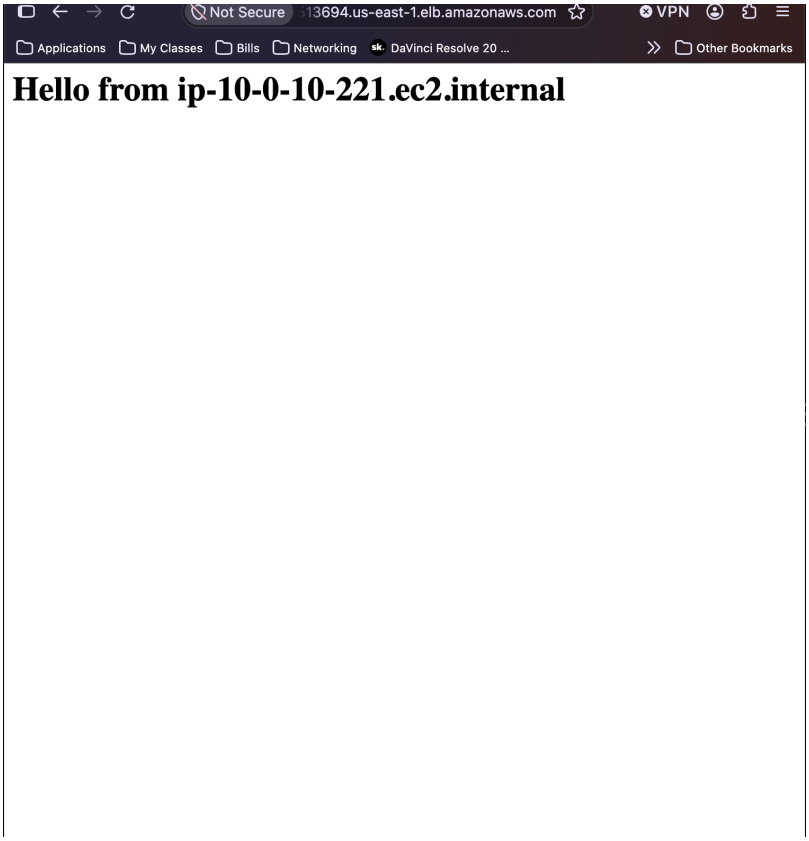
Configuration: Engine 8.4.11, db.t4g.micro 2 vCPU 1 GB, Storage 20 GiB gp2, Encryption Enabled aws/rds, Created Nov 10 2026 22:22, Deletion protection Disabled, Multi-AZ No. DB name blank, no p1db initial DB. Storage autoscaling Enabled max 1000 GiB, Enhanced Monitoring Enabled, both left on by console defaults.



File: screenshots/database/step10-rds-p1-db-configuration.png

E51 - 2026-11-10 22:32 - step10-alb-test-with-db.png - Part B ALB check passed

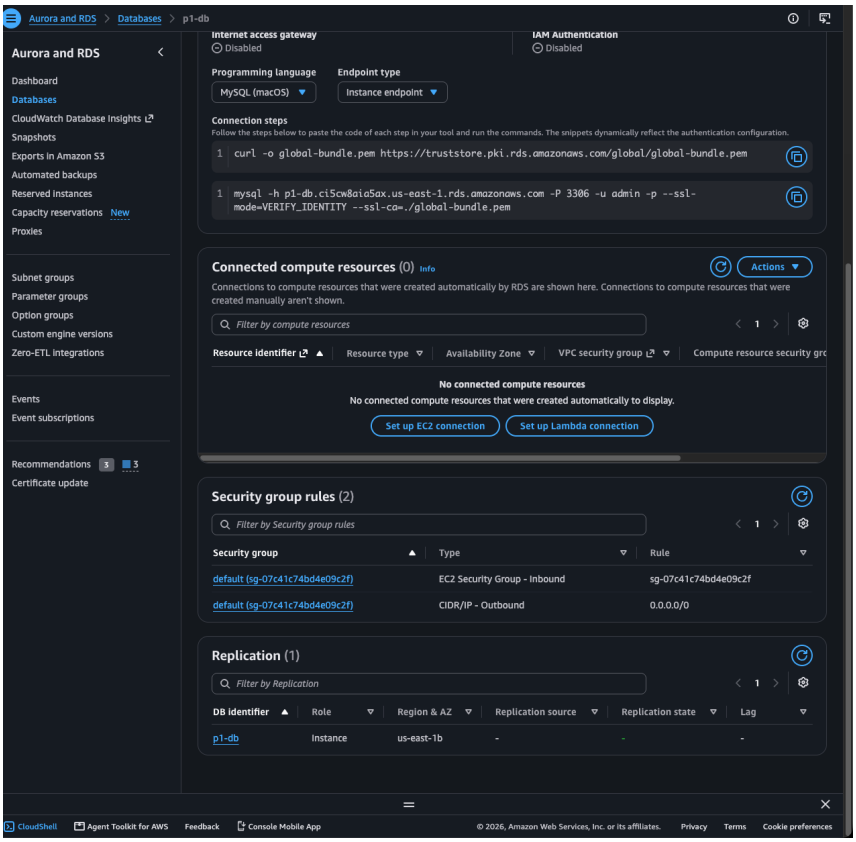
Browser, ALB URL, page: Hello from ip-10-0-10-221.ec2.internal. ALB still serves traffic with p1-db present. This is the last successful screenshot, and the end point of the build at Haileab's direction.



File: screenshots/load-balancer/step10-alb-test-with-db.png

E52 - 2026-11-10 22:33 - step10-rds-p1-db-security-group-wrong-default.png - SG wrong - not fixed

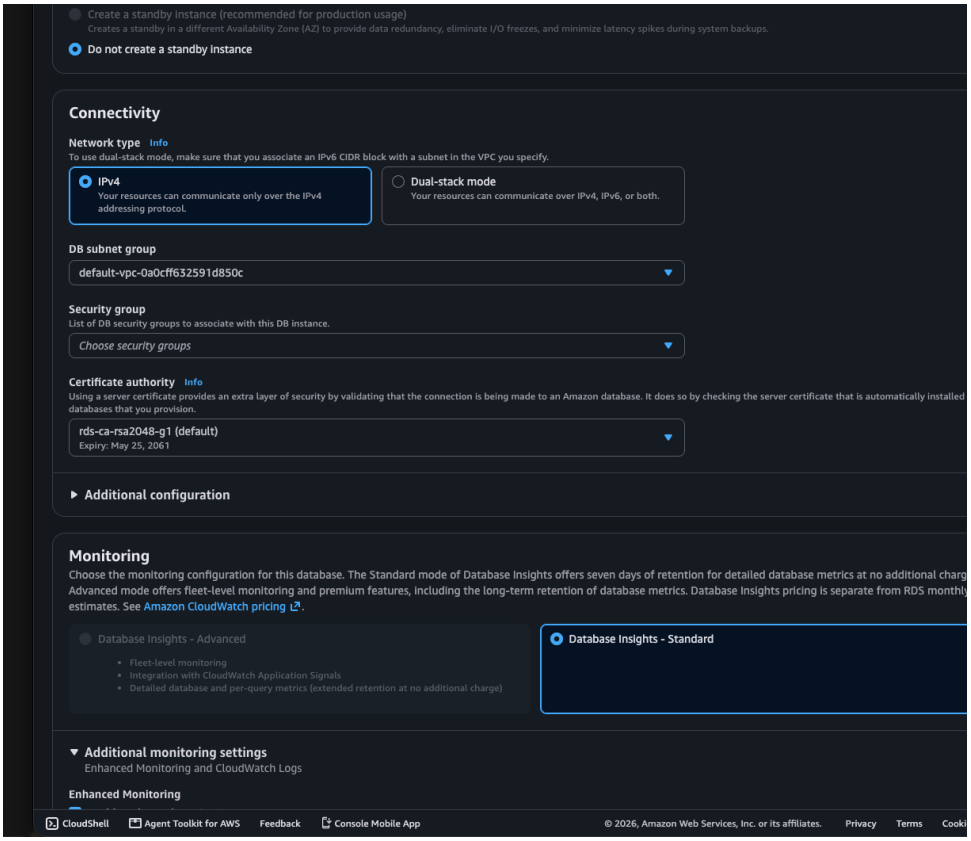
Security group rules (2) on p1-db: default (sg-07c41c74bd4e09c2f) inbound and outbound, not p1-sg-db sg-063e10e3231772a8b. App tier cannot reach DB on 3306 through this group.



File: screenshots/database/step10-rds-p1-db-security-group-wrong-default.png

E53 - 2026-11-10 22:36 - step10-modify-db-subnet-group-default-vpc.png - VPC wrong - not fixed

Modify DB page, DB subnet group: default-vpc-0a0cff632591d850c. p1-db was created in the DEFAULT VPC, not p1-vpc, in default subnets. VPC/subnet group cannot be changed by Modify. Fix is delete and recreate in p1-vpc, not done at Haileab's direction.



File: screenshots/database/step10-modify-db-subnet-group-default-vpc.png